Supplementary material

Cryptic genomic variation in a panmictic squid: an ancient inversion supergene and genome-wide signatures of positive selection in *Illex illecebrosus*

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Supplementary methods

Read processing and variant calling. Raw reads for all 633 individuals were downloaded from ENA project PRJEB101095 and processed with grenepipe (Czech and Exposito-Alonso, 2022). Reads were trimmed with fastp (Chen et al., 2018) and mapped to the `Illex_F24` primary assembly with BWA-MEM2 (Vasimuddin et al., 2019), and duplicates were removed with Picard MarkDuplicates. Each individual was genotyped with GATK HaplotypeCaller in gVCF mode, and the gVCFs were jointly genotyped with all sites retained (Poplin et al., 2018). Variant records were annotated with snpEff and soft-filtered with bcftools (Danecek et al., 2021). A site was removed if its quality was below 20, its root-mean-square mapping quality was below 40, or its strand-bias score (FS) exceeded 60. It was also removed if it was missing in half or more of the individuals, if it lay in a GC-corrected high-depth window identified with mosdepth, or if it lay outside the accessibility mask. Sites at intermediate allele frequency (0.1 to 0.9) with a strong excess of heterozygotes (ExcHet < 0.001) were flagged as likely paralogous and removed. Only sites passing every filter were used. The accessibility mask is the 1.43 Gb of the 3.47-Gb assembly that lies outside repeats annotated with Earl Grey. Mappability was computed with GenMap 1.3.0 (Pockrandt et al., 2020) for 150-mers with up to two mismatches, matching the 151-bp reads. Across the assembly 14.7% of positions are not uniquely mappable, but within the accessibility mask only 3.8% (53.7 Mb) are, because the repeat annotation already removes nearly all multi-copy sequence. Non-unique sequence is concentrated six-fold in the 22 anomalous-diversity blocks (23% of their accessible sequence), and outside them it is 2.7% of accessible sequence. For the sweep scan and the gene-set analyses, the mask was extended by the 22 anomalous-diversity blocks (129 Mb, Supplementary Table S9), in which collapsed duplications and sequence duplicated within the assembly make genotype-based statistics unreliable.

Diversity and demography. All site-frequency spectra and diversity statistics were computed with ANGSD 0.940 (Korneliussen et al., 2014) directly from the deduplicated BAM alignments of the 633 individuals, not from the called genotypes, so that the uncertainty of genotypes at about 9-fold depth is carried through as genotype likelihoods. Each chromosome was run separately with the options `-doSaf 1 -doCounts 1 -GL 1 -minQ 20 -minMapQ 20 -remove_bads 1 -only_proper_pairs 1 -minInd 1 -setMinDepthInd 1`, restricted with `-sites` to the accessibility mask and with the reference supplied as both `-ref` and `-anc`. `-GL 1` is the SAMtools

genotype-likelihood model, and because the reference is not an ancestral sequence all spectra are folded. The folded spectrum was estimated with `realSFS -fold 1 -tole 1e-8 -maxIter 200`, per-site thetas with `realSFS saf2theta -fold 1`, and π, Watterson's θ and Tajima's D with `thetaStat do_stat`, per chromosome and in 50-kb windows (10-kb step genome-wide, 50-kb step per division). The genome-wide folded spectrum pooled across chromosomes is the input to the demographic models. Per-division statistics used the same options on at most 25 individuals per division (all individuals in 3O, 4S and 4T), and the Z was run in the 330 confidently sexed males. Per-karyotype spectra and genotype-likelihood F_{ST} between arrangements used the 2a/2a and 2+/2+ homozygotes as separate sample sets. Stairway Plot 2, moments (a two-epoch growth model fitted to the folded spectrum, with Godambe confidence intervals), and GONE2 were fitted as described in Supplementary Table S1, and the moments model was checked by forward simulation, which reproduced the observed π and Tajima's D.

Genotype-based statistics. Statistics that need individual genotypes were computed with pg_gpu (Pope et al., 2026) from the filtered VCF. Windowed π, d_{xy} and Hudson F_{ST} between arrangements used biallelic SNPs in 100-kb windows, with the 2a/2a ($n = 254$) and 2+/2+ ($n = 95$) homozygotes as the two populations and missing genotypes included pairwise. The between-division F_{ST} matrix (Hudson), the linkage-disequilibrium triangles (genotype r^2 among common SNPs), and the principal-component analyses were computed the same way. The per-SNP geographic F_{ST} scans (Supplementary Figs. S2, S19) used plink2 (Chang et al., 2015) on every biallelic SNP with a minor-allele frequency above 0.05 in the pooled 632 individuals from the ten divisions with more than one sample (division 3N, with a single individual, was excluded), giving 20,276,930 SNPs. Per-division allele counts were converted to Hudson F_{ST} for each of the 45 division pairs (Bhatia et al., 2013), at sites genotyped in at least 40 individuals in that pair and with at least two alleles in each division. The per-SNP value is the ratio of averages over informative pairs (the sum of the numerators over the sum of the denominators), without clipping, keeping sites informative in at least 20 of the 45 pairs. The null was built by shuffling division labels among the 632 individuals 20 times (division sizes kept) and recomputing F_{ST} at the same SNPs in the same way. Thresholds are quantiles of the pooled null over the autosomes outside the chromosome-2 inversion (3.996×10^8 values), and the expected count above each threshold is the number of SNPs times the null tail rate.

The inversion. Karyotypes were called from the first principal component of a local PCA across chromosome 2 with a three-component Gaussian mixture (95.3% of individuals assigned with posterior probability above 0.95). Breakpoints were placed at the step change in between-arrangement differentiation and confirmed by a strand flip in the AnchorWave alignment to *I. coindetii*. The arrangement carried by each haplotype of the reference individual was read from minimap2 2.29 (`-x asm5`) alignments of chromosome 2 of the two haplotype assemblies to each other and to the primary assembly, and the orientation of each haplotype relative to *I. coindetii* from alignments to the homologous *I. coindetii* chromosome (OZ346549.1, `-x asm20`). Alignment blocks of at least 1 Mb were ordered along each *Illex illecebrosus* haplotype, and polarity was read from which arrangement keeps the order and adjacency of *I. coindetii* sequence across the rearranged segment (Supplementary Table S3). The same comparison was made with the chromosome-level assembly of *Sthenoteuthis oualaniensis* (Li et al., 2022), a member of the same family. Its chromosome-2 homolog, HiC_scaffold_1, is the scaffold that receives 265 of the 319 *Illex illecebrosus* chromosome-2 genes transferred with LiftOn (Chao et al., 2024), and chromosome 2 of each haplotype assembly was aligned to it with minimap2 (`-x asm20`). Because the two genera differ by many small rearrangements along the chromosome, polarity was read at each breakpoint from which flanking

segments are joined, using alignments of at least 5 kb with mapping quality of at least 5 together with the order of the transferred genes. Allele sharing with *I. coindetii* was also scored at the 13,998 sites in the inversion where allele frequency differs by at least 0.7 between 2a/2a and 2+/2+ homozygotes, with the *I. coindetii* base read from the AnchorWave alignment of *I. coindetii* to the primary assembly, which carries 2+ in the inversion. Where the *I. coindetii* base was callable (13,582 sites), the majority allele of 2a matches it at 25.0% of sites, so 2+ carries the *I. coindetii* allele at most diagnostic sites. The 2a matches are consistent with variation that was segregating in the ancestral population and was captured by the inversion when it formed, possibly in more than one capture event, which would give the soft start discussed in the main text. Chromosome 2 of both outgroups is rearranged in this region, so the arrangement of the common ancestor away from the breakpoints cannot be determined. It may have been an intermediate rearranged form, and the polarity used here is the reading on which *I. coindetii* and *S. oualaniensis* agree at both breakpoints. Restricted to the 4,175 sites where *I. coindetii* differs from the reference base, 2a carries the *I. coindetii* allele at 63.0% (2,631) and 2+ at 37.0% (1,544), but this subset is selected for sites at which the reference, a 2+ haplotype, carries a non-*I. coindetii* allele, so it is biased toward 2a matches. The age and frequency trajectory were fitted with msinv as described in the next paragraphs. Unannotated, non-repeat sequence in the inversion was searched with DIAMOND `blastx` against seven cephalopod proteomes, with transposable elements from Earl Grey (Baril et al., 2024) and transcripts from StringTie2 (Kovaka et al., 2019) as supporting evidence. This recovered 14 likely protein-coding loci that the annotation lacks. They are provided as `protein_match` features in version 3 of the annotation and are not counted among the 89 gene models of the inversion. Fixed differences were sites with an allele-frequency difference of at least 0.9 between 2a/2a and 2+/2+ homozygotes. Gene-ontology enrichment of the inversion genes used Fisher's exact test against the annotated-gene background in all three ontologies, compared with 500 random gene sets of equal size, and gene density was compared with all autosomal windows of the same length (0.5-Mb steps). The two drift calculations are not separate estimates. They are analytic diffusion calculations that take the plateau frequency (0.22–0.38) and the duration of the rise from the msinv grid-search fit and N_e from the fitted demography, and ask whether drift alone could produce that trajectory. The attainment calculation used the expected time for a new neutral mutation to reach frequency x, conditional on reaching it, $(4N_e/x)[x + (1 - x)\ln(1 - x)]$ ($0.476\,N_e$ at $x = 0.22$ and $0.880\,N_e$ at $x = 0.38$), and the rise bound used the drift standard deviation $\sqrt{p(1-p)\,t/(2N_e)}$ with N_e from the fitted demography over the rise. The load test compared the zero-fold fraction among the most differentiated coding sites (genotype-likelihood F_{ST}, 40 individuals per karyotype) with all coding sites in the inversion by Fisher's exact test, and per-transcript d_{xy} was normalised to its 100-kb window and compared with collinear control transcripts.

Inversion coalescent model. The age and frequency history of the inversion were fitted with msinv v0.3.5 (`https://github.com/stsmall/msinv`, commit `35412bf`), a coalescent simulator with recombination that writes tree sequences. It uses the sequentially Markov approximation (McVean and Cardin, 2005), extended to two karyotype classes as in Peischl et al. (2013), in which two lineages coalesce only where they share ancestral material. This approximation reproduces the expected coalescence time at each site, which is all the fitted statistics use. The model code, fitting scripts, and every result quoted here are in the `illex/` and `results/illex/` directories of that repository. Inside the inversion every lineage carries a karyotype label, ancestral (S, the 2+ arrangement) or derived (I, the 2a arrangement). Two lineages of the same class coalesce at rate $1/[2N(t)\,p_c(t)]$ per generation, where $N(t)$ is the effective size and $p_c(t)$ the frequency of their class, and recombination within the inversion occurs only between chromosomes of the same arrange-

ment, at rate $r\,p_c(t)$. Lineages of different classes cannot coalesce while the barrier exists. msinv can also exchange short tracts between arrangements by gene flux, with the position-dependent rate of Peischl et al. (2013), which rises from zero at the breakpoints toward the centre of the inversion. In these fits the flux rate was set to 10^{-15} per site per generation, which is effectively zero, because divergence between arrangements is flat from the breakpoints to the centre of the inversion (edge-to-centre ratio of d_{xy}/π 0.999). Backward in time, the barrier is removed at t_{inv}, when the derived lineages rejoin the ancestral class. t_{inv} is therefore the age of the recombination barrier, and the model does not impose a single-copy origin or a separate rise of the new arrangement.

Each simulation was a 37,148-bp sequence ($4N_0rL = 2000$), consisting of a 29,718-bp inversion with 3,715 bp of collinear sequence on each side, and sampled 100 chromosomes of each arrangement. The demography was the fitted moments model, with exponential growth from 547,928 to 6,808,096 over the last 769,519 generations and a constant size before that. The recombination rate was $r = 1.977 \times 10^{-9}$ per site per generation, the rate of the collinear part of chromosome 2 in the recombination map, and $\mu = 3 \times 10^{-9}$. Statistics were computed from branch lengths within the inversion, so the fitted ratios do not depend on μ, and μ sets only the absolute time scale through the demography.

Frequency trajectory. The frequency of each arrangement followed a deterministic trajectory, with neutrality within each arrangement. The fit was parameterised by the frequency of 2+, which is one minus that of 2a, and because the model treats the two classes identically apart from their labels it is given here for the derived arrangement 2a. Going back from the present, the frequency is $p_{\mathrm{now}} = 0.626$ until t_{end}, the time at which the rise ended. Before that it rises from a plateau p_{hist} over a duration t_{rise}, and it stays at p_{hist} from the start of the rise back to t_{inv} (t_{end} and t_{rise} are `t_decline` and `t_fall` in the code). Forward in time, the rise follows $dp/dt = (s/p^*)\,p(1-p)(p^*-p)$ toward the equilibrium $p^* = p_{\mathrm{now}}$, the form expected under overdominance, with s solved so that the frequency comes within 0.01% of p_{now} in exactly t_{rise}. Most of the rise therefore happens early, and for a 100-ky rise from 0.26 the frequency reaches 0.60 within the first 28 ky. The value of s summarises the shape of the fitted trajectory. It is not a fitted selective mechanism.

Fitted statistics. Three statistics were fitted. The diversity ratio π_S/π_I (2+ over 2a) and the divergence ratio d_{xy}/π_S were computed for each replicate and averaged over replicates. Their empirical values, 1.356 ± 0.048 and 1.385 ± 0.021, come from 254 2a/2a and 95 2+/2+ individuals over the inversion (189 windows of 100 kb between 60.5 and 79.5 Mb, pg_gpu), with standard errors from a delete-one jackknife over 1-Mb blocks. The third statistic is the ratio between arrangements (2+ over 2a) of the singleton-class fraction of the site-frequency spectrum, which in the simulations is the first bin of the folded per-arrangement spectrum projected to 20 chromosomes. Its empirical value from per-arrangement ANGSD spectra is 0.8256 ± 0.0076 in the inversion and 1.0161 in a collinear control region (chr2:10–30 Mb), and the fitted target was the control-calibrated ratio 0.8125 ± 0.0076.

Fitting and profiles. The trajectory was fitted by grid search. Coarse exploratory grids spanned t_{inv} from 200 ky to 6 My and p_{hist} from 0.12 to 0.55. The final grid has 2,990 cells of 48 replicates, concentrated around the best fit, with t_{inv} from 700 to 1,000 ky (25-ky steps from 775 ky), p_{hist} from 0.22 to 0.44 in steps of 0.02 and 0.18, t_{end} of 0, 25, 50, 75, 100, 150, 175, 200 and 250 ka, and t_{rise} of 50, 100, 150, 200, 400 and 600 ky. Each cell was scored with the covariance-weighted distance $D = (\hat{x} - x)^{\top}\Sigma^{-1}(\hat{x} - x)$, where $\hat{x}$ holds the simulated means of the three statistics and x their empirical values. Σ is the sum of two parts. The measurement part has the jackknife

standard errors, the correlation between π_I/π_S and d_{xy}/π_I across 20 1-Mb blocks of the inversion (-0.71), and an independent singleton ratio, which comes from a separate estimator. The trajectory part is the variance that a single realised history adds and that does not average out along the inversion. It was estimated at a cell with a slow rise (825 ky, plateau 0.26, a 200-ky rise ending 100 ka) as the covariance of the per-replicate statistics along 384 Wright–Fisher trajectories, with selection of the fitted strength and the fitted $N(t)$, minus their covariance along 384 replicates of the deterministic trajectory. Its standard deviations are 0.126, 0.013 and 0.026 for the three statistics, against measurement standard errors of 0.048, 0.021 and 0.0076. A weaker-drift cell with a 100-ky rise gives 0.046, 0.003 and 0.011 and a plateau interval of 0.24–0.32, so the intervals below use the larger estimate. Under drift the mean π_S/π_I is 0.03–0.05 higher and the mean singleton ratio about 0.01 lower than along the deterministic trajectory, shifts within the trajectory variance. A version in which 2+ starts from a single copy at t_{inv} and rises under selection to a plateau of 0.62–0.78 gives $\pi_S/\pi_I = 0.69$, $d_{xy}/\pi_S = 2.72$ and a singleton ratio of 1.10 ($\chi^2 \approx 5{,}500$ against the targets), so under the reverse polarity t_{inv} would be the age of the barrier and not of a single-copy origin. A single-copy origin of 2a rising to a plateau of 0.22–0.38 was not part of this grid.

The best cell is $t_{inv} = 825$ ky and $p_{hist} = 0.28$, with a 50-ky rise that ended 250 ka ($D = 0.19$). Profiles of ΔD, in which the other parameters were re-optimised over the grid, bound each parameter (Supplementary Fig. S14). The ΔD cutoffs were calibrated by a parametric bootstrap. At the best cell, 400 pseudo-observed datasets were drawn by adding noise with covariance Σ to the simulated statistics, the statistics of every cell were recomputed from a resample of its own replicates, and each pseudo-dataset was refitted on the whole grid with the same D. The cutoff for each parameter is the 95th percentile of its profile ΔD at the true value, 3.2 for the barrier age, 3.4 for the plateau, 2.3 for the end of the rise and 2.3 for its duration, and the grid values within these cutoffs form the calibrated 95% sets. The barrier age is 800–850 ky, with ΔD of 5.4 at 775 ky, 12.8 at 750 ky, 34 at 700 ky, 4.8 at 875 ky, 11.6 at 900 ky and 86 at 1,000 ky. The plateau is 0.22–0.38, with ΔD of 3.2 at 0.38, 4.3 at 0.40, 6.0 at 0.44 and 7.9 at 0.18. Rise durations of 50 to 200 ky are not excluded ($\Delta D \leq 0.1$), while 400 and 600 ky give ΔD of 5.3 and 9.4. The rise ended 100–250 ka, 250 ka being the oldest end tested, while ends at 75, 50, 25 and 0 ka give ΔD of 6.5, 3.7, 8.3 and 16.8. Across the plateaus and rise durations that are not excluded the trajectory implies s from 1.2×10^{-4} to 5×10^{-4} ($N_e s$ of about 250 to 2,200 at the middle of the rise), and $s \approx 2.3 \times 10^{-4}$ for a 100-ky rise. These values are those of the 2+ parameterisation multiplied by 0.626/0.374, the factor that converts s/p^* between the two labellings. The coverage of these sets was tested with every grid cell in turn as the truth, using 400 pseudo-datasets per cell and the same cutoffs. Across truths the calibrated sets contain the true barrier age in a median of 95% of pseudo-datasets, the plateau in 95%, the end of the rise in 98% and its duration in 99% (10th percentiles 91–93%). Of the 18 cells at least 100 ky younger than the best fit, 2.0% of pseudo-datasets (at most 4.5% for any cell) are recovered within 50 ky of the best-fitting age, so a younger inversion with a different trajectory is not mistaken for the fitted one. The grid, the trajectory variance, the bootstrap and the recovery are `illex/scripts/refit_decline.py`, `stochastic_check.py` and `param_recovery.py` in the msinv repository at commit `a9f92bb`. The bootstrap holds the demography, mutation rate and recombination rate at their fitted values, so these are 95% intervals conditional on the model.

Simulated length and validation. The fits used a 29.7-kb simulated inversion, and the fitted values were applied to the 19-Mb inversion. At the best-fitting trajectory, the three fitted statistics are nearly constant over a 25-fold range of simulated lengths (3.0 to 74 kb of inversion, 120 replicates per length). Extrapolating the 95% confidence limits of their slopes from the fitted length to 19 Mb bounds the change at 4.6% for π_I/π_S, 1.7% for d_{xy}/π_I and 2.8% for the singleton ratio. With an

inversion, under constant population size and without gene flux, the mean pairwise coalescence times within each arrangement, between arrangements, and in the collinear flanks match their exact expectations for arrangement frequencies of 0.2 to 0.74 and barrier ages of $0.25N$ to $4N$ generations (24 comparisons, 200 replicates each, all within 2.7 standard errors). The current version of msinv (commit `e53f0c6`) also agrees with forward simulations in SLiM (Haller et al., 2026) that share each replicate's realised frequency trajectory, in four scenarios covering a short and a long barrier, low recombination, and an arrangement held near 0.74 by overdominance (200 replicates each). Across these scenarios the mean coalescence times agree within 4%, the medians of both fitted ratios agree within 4%, and $|d| < 0.2$ for every statistic. At the best-fitting trajectory the fitted statistics from this version and from v0.3.5 differ by at most 0.3%. The fits use only the means of the fitted statistics, which at the fitted length vary by about 2% among replicates. Statistics that depend on the correlation of genealogies along the sequence are not used.

Recombination map. ReLERNN (Adrion et al., 2020) was run in unphased mode on biallelic SNPs with at most 50% missing genotypes. Individuals with no genotype calls on a chromosome cannot enter the unphased model, so one of the 330 confidently sexed males was removed and 329 were used. The autosomal network was trained on chromosomes 1, 13 and 45 with 12,000 coalescent simulations (with 1,000 each for validation and testing) under the Stairway Plot 2 demography with $\mu = 3 \times 10^{-9}$, an upper bound of 4 on the ratio of ρ to θ, and 6-kb windows, masked by the inaccessible regions and the 22 π-elevated blocks with a window-level masking threshold of 0.9. It was trained for 50 epochs and applied to every autosome except chromosome 2, in windows of 3–6 kb. The Z network was trained in the same way in 30-kb windows, with the Z accessibility mask and the effective sizes of the demography multiplied by 0.60, the Z-to-autosome diversity ratio. Predictions were corrected with 200 parametric-bootstrap replicates in each of 20 rate slices. The map pools all karyotypes and reads linkage disequilibrium within windows of a few kilobases, so it does not resolve the barrier between chromosome-2 arrangements, which is shown instead by long-range linkage disequilibrium (Supplementary Figs. S4, S24).

Z recombination and sex-biased mutation. Linkage disequilibrium on the Z and on the autosomes was compared in the 330 ZZ males, the only individuals that carry two Z copies. Two-locus statistics (D^2 and π_2, unbiased for unphased genotypes (Ragsdale and Gravel, 2020)) were computed with pg_gpu (Pope et al., 2026) on accessible biallelic SNPs in 2-Mb windows, 86 on the autosomes (at 10 and 40 Mb of each autosome, excluding the chromosome-2 inversion and the anomalous-diversity blocks) and nine spaced every 4 Mb along the Z, with SNP pairs binned by distance from 0 to 5 kb. The decay of $\sigma_D^2 = D^2/\pi_2$ with distance was fitted to the expectation of Ohta and Kimura (1971) plus a constant sample-size term, giving the population-scaled rate $\rho = 4N_e r$ per base for each chromosome class. The ratio ρ_Z/ρ_A was also estimated without a model, as the distance rescaling that best superimposes the two decay curves. Confidence intervals come from 200 bootstrap resamples of windows. Because the Z recombines only in males and spends two thirds of its time in them, $\rho_Z/\rho_A = (N_{e,Z}/N_{e,A})\,(4/3)\,r_m/(r_m + r_f)$, where r_m and r_f are the male and female per-generation rates and the male rate is taken to be the same on the Z and the autosomes. With $N_{e,Z}/N_{e,A} = 3/4$, equal recombination in the two sexes predicts $\rho_Z/\rho_A = 0.5$ and achiasmate female meiosis predicts 1 (Coffing et al., 2026).

Sex-biased mutation was estimated from synonymous divergence between *Illex illecebrosus* and *S. oualaniensis*, whose chromosome 46 is syntenic with the *Illex illecebrosus* Z (Small et al., 2026). The *Illex illecebrosus* annotation was transferred to the *S. oualaniensis* assembly with LiftOn (Chao et al., 2024), one transcript per gene was aligned codon by codon with MACSE 2.07 (Ranwez et al.,

2018), which allows frameshifts in the transferred sequence, and codons with gaps, frameshifts, ambiguous bases or stop codons were removed. Synonymous and nonsynonymous divergence (d_S, d_N) were computed per gene by the method of Nei and Gojobori (1986) with a Jukes–Cantor correction. Genes in the chromosome-2 inversion or the anomalous-diversity blocks were excluded, leaving 8,649 autosomal and 109 Z-linked genes with $d_S > 0$. The male-to-female mutation ratio $\alpha = \mu_m/\mu_f$ was obtained from the ratio R of mean Z to mean autosomal d_S as $\alpha = (3R-2)/(4-3R)$ (Miyata et al., 1987), as in Coffing et al. (2025) and Coffing et al. (2026). Z-linked genes carry more gene-body CpG methylation than autosomal genes in *Illex illecebrosus*, most strongly in females (Small et al., 2026). If this extends to the germline, methylation-associated mutation raises the Z rate and biases α upward, so the female bias is if anything underestimated. To separate mutation from selective constraint and base composition, R was recomputed with autosomal genes matched to the Z genes by deciles of d_N/d_S and of third-position GC content (GC3), and with genes restricted to those above the autosomal median d_N/d_S. Confidence intervals come from 2,000 bootstrap resamples of genes within strata.

Linkage disequilibrium decays at the same rate on the Z as on the autosomes, $\rho_Z/\rho_A = 1.04$ (95% CI 0.88–1.28) from the decay fit and 1.08 (0.78–1.67) from curve rescaling, with $\rho_A = 0.026$ per base (Supplementary Table S10). Synonymous divergence is lower on the Z ($d_S = 0.298$ vs 0.358, Mann–Whitney $p = 0.009$), giving $R = 0.83$ and $\alpha = 0.33$ (95% CI 0.10–0.70), a female mutation rate about three times the male rate. Z-linked genes are slightly more constrained (median d_N/d_S 0.082 vs 0.113, $p = 0.048$) and lower in GC3 (0.36 vs 0.42, $p < 10^{-16}$). Matching on d_N/d_S leaves α unchanged (0.36), and matching on GC3 raises it to 0.43, or 0.49 when combined with d_N/d_S, so the female bias holds under each control, with the upper confidence limit reaching 1 only under joint matching. Because R estimates μ_Z/μ_A, the ratio of male Z to autosomal diversity ($\pi_Z/\pi_A = 0.00563/0.0093 = 0.61$) implies $N_{e,Z}/N_{e,A} = 0.68$–0.76 across these estimates, close to the 3/4 expected under an equal breeding sex ratio. Combined with the linkage-disequilibrium ratio, this gives $r_m/(r_m+r_f) \approx 1.02$–$1.14$, consistent with achiasmate female meiosis (Coffing et al., 2026) and inconsistent with equal recombination in the two sexes, which predicts 0.5.

Selective-sweep scan. diploS/HIC used 1.1-Mb feature windows of eleven 100-kb sub-windows and twelve diploid statistics per sub-window. It was trained on hard and soft sweeps simulated in SLiM 5.2 and recapitated with msprime, and on neutral msprime simulations, all under the fitted growth demography, at the empirical sample size, and with the empirical per-individual missing-data structure. The final model classified held-out simulations with 81% accuracy (Supplementary Fig. S7). Applied to the genome it assigns 1% of windows to the neutral class and 91% to the linked classes, because real neutral windows carry three to eight times more spatial heterogeneity in diversity among sub-windows than the simulations. We therefore used it as an outlier detector. The sweep score $p_{\text{hard}} + p_{\text{soft}}$ was ranked within strata of callable fraction (quintiles) and coding density (zero or not), and windows in the top 1% of their stratum were outliers. RAiSD and SweepFinder2 statistics were ranked the same way, and a region was concordant when a diploS/HIC outlier was supported by either. Tier-1 calls came from a hidden Markov model with neutral, linked and sweep-centre states, whose emissions are the prior-corrected class posteriors and whose dwell lengths follow the ∼400-kb sweep footprint scaled by the recombination map. Calibrating this decoder by simulation proved infeasible, because at this population size the simulated ancestral recombination graphs carry about 3.4×10^5 marginal trees per 2.5 Mb. Its settings are therefore design-based and were checked against the concordant regions (Supplementary Fig. S13). The background-selection screen compared each call's π, from the ANGSD windows it overlaps, with the value predicted from the fraction of coding sites within 1 Mb of its midpoint, $\hat{\pi} = 0.010593 - 0.165473\, f_{\text{CDS}}$, the genome-

wide linear relationship between the two. A call was retained when $\pi - \hat{\pi} \leq -0.003$, i.e. when its diversity is at least 0.003 below the background-selection expectation. Retained calls with f_{CDS} above the genome median are those well below the expectation, and those below it are low-diversity calls in gene-poor sequence where background selection is weak. Gene-ontology enrichment used genes with EnTAP (Hart et al., 2020) annotations that overlap a call, against all annotated genes outside the Z and the masked blocks.

Anomalous-diversity blocks. Blocks were called where the rolling median of 10-kb π exceeded twice the chromosome median over at least 1 Mb. Each was characterised by per-division π, between-division F_{ST}, local PCA, and genotype linkage disequilibrium, and by read depth relative to 5-Mb flanks in all 633 individuals (samtools bedcov at mapping quality 0 and 20). Long-read depth and heterozygous-site allele fractions came from PacBio HiFi reads of the assembly individual and a second individual aligned to the assembly, and genotype-likelihood π was recomputed at mapping quality 60. Blocks with population depth at least 1.15 times their flanks were classed as collapsed duplications and those at 0.90 or less as under-mapped, and the fraction of each block's accessible sequence that is not uniquely mappable was taken from GenMap (Supplementary Table S9).

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Supplementary figures

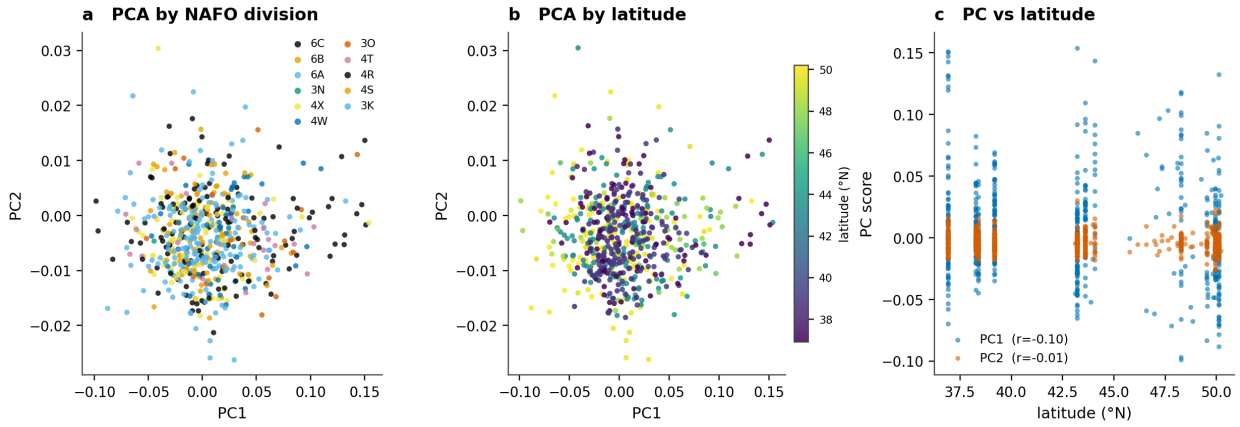


Figure S1: **The autosomal-PCA spread is not geographic structure.** Genome-wide PCA (autosomes, chromosome 2, chromosome 42 and the Z excluded), $n = 621$ individuals. (a) Coloured by NAFO sampling division and (b) by latitude, the divisions and the latitudinal gradient are fully intermixed across PC space, with no clustering by origin. (c) PC scores plotted against latitude show no relationship (Pearson $r = -0.10$ for PC1, $r = -0.01$ for PC2). The dispersion along the minor PC axis in main-text Fig. 2 is carried by a handful of individuals ($\sim$9 samples account for $>$80% of the PC2 variance) and tracks neither division nor latitude nor sequencing coverage. A KING-robust screen finds no cryptic relatedness among them, so it is a minor-axis artifact rather than population subdivision, consistent with genome-wide $F_{ST} \approx 0$.

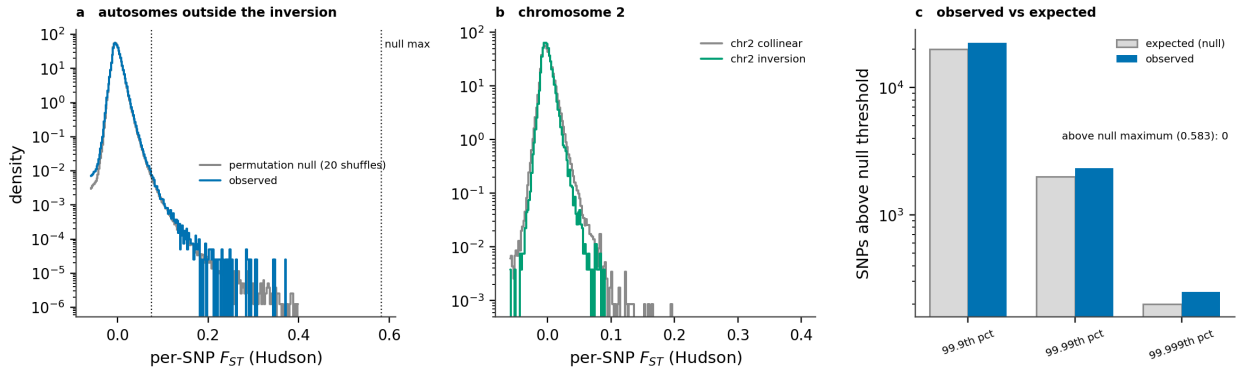


Figure S2: **No geographic differentiation beyond genetic drift and no local adaptation signal.** Per-SNP Hudson F_{ST} across all 45 NAFO-division pairs for every biallelic SNP with minor-allele frequency above 0.05 ($\geq$40 genotyped individuals per pair, ratio of averages over $\geq$20 informative pairs). **(a)** On the autosomes outside the inversion (19,984,027 SNPs), the empirical distribution (blue) matches the permutation null (grey), built by shuffling division labels among individuals 20 times, which keeps genotype calling, missingness, the allele-frequency spectrum and linkage. Empirical maximum $F_{\mathrm{ST}} = 0.371$ against a null maximum of 0.583. **(b)** On chromosome 2, the inverted region (green) is no more geographically differentiated than the collinear remainder, so the inversion is non-clinal and its $F_{\mathrm{ST}} \approx 0.37$ is between karyotypes, not geography. Its upper tail is lighter than the null (66 SNPs above the null 99.9th percentile against 134 expected). **(c)** SNPs above the null 99.9th, 99.99th and 99.999th percentiles (22,562, 2,319 and 249) are within the permutation range ($19{,}979 \pm 2{,}154$, $1{,}998 \pm 352$ and 200 ± 54, permutation $p = 0.19$, 0.24 and 0.29), and no SNP exceeds the null maximum, so a scan for local adaptation that the diversity-based sweep tests could miss returns no candidates. Because the minor-allele-frequency filter is applied to the pooled 632 individuals, a variant private to one small division can fall below it (a SNP fixed only in 3O, with 16 individuals, has a pooled frequency of 2.5%), so the scan rules out strongly differentiated SNPs that are common overall, but not private alleles in the smallest divisions.

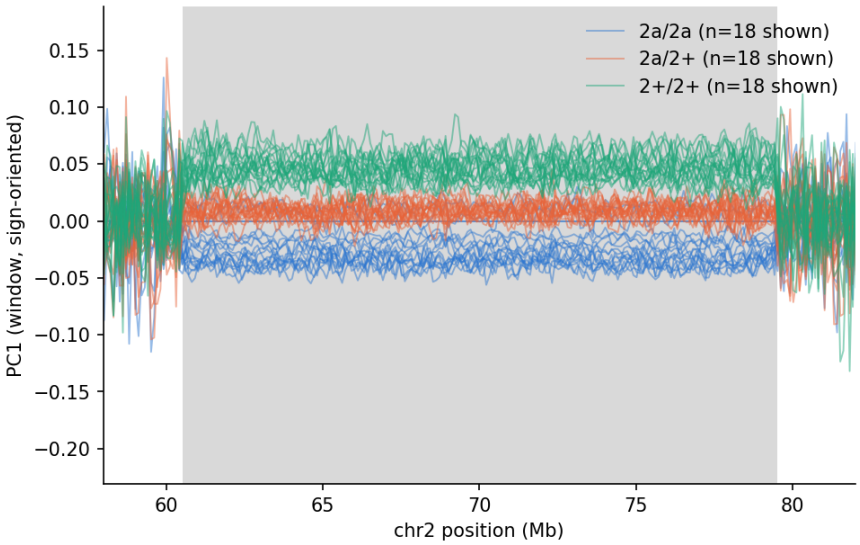


Figure S3: **Windowed PCA across chromosome 2** localising the differentiated block.

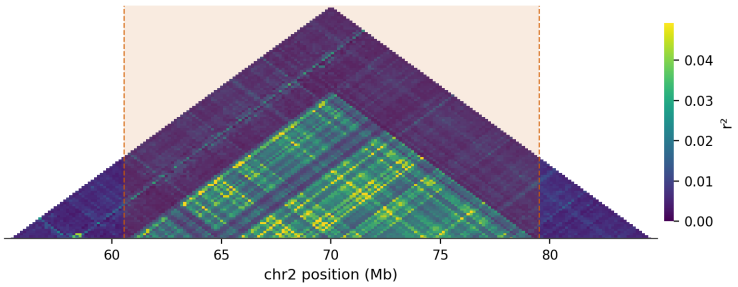


Figure S4: **Linkage disequilibrium across the chromosome-2 inversion.** Mean genotype r^2 between SNPs, binned along chr2:55–85 Mb, pooling all 633 individuals. Dashed lines and shading mark the breakpoints (60.54 and 79.50 Mb). A block of elevated linkage disequilibrium spans exactly the inversion and stops at both breakpoints, while the flanking collinear sequence shows only background values. Mean r^2 between SNPs more than 1 Mb apart is 3.5 times higher inside the inversion than in its flanks (0.0056 vs 0.0016, Supplementary Fig. S24). This long-range association arises because the few near-fixed differences between arrangements are all correlated with karyotype, and it is the direct signature of suppressed recombination between arrangements. Recombination within arrangements is not suppressed at the scale of the 3–6-kb windows used by the recombination map, which is why the inversion is shown here rather than in Supplementary Fig. S12.

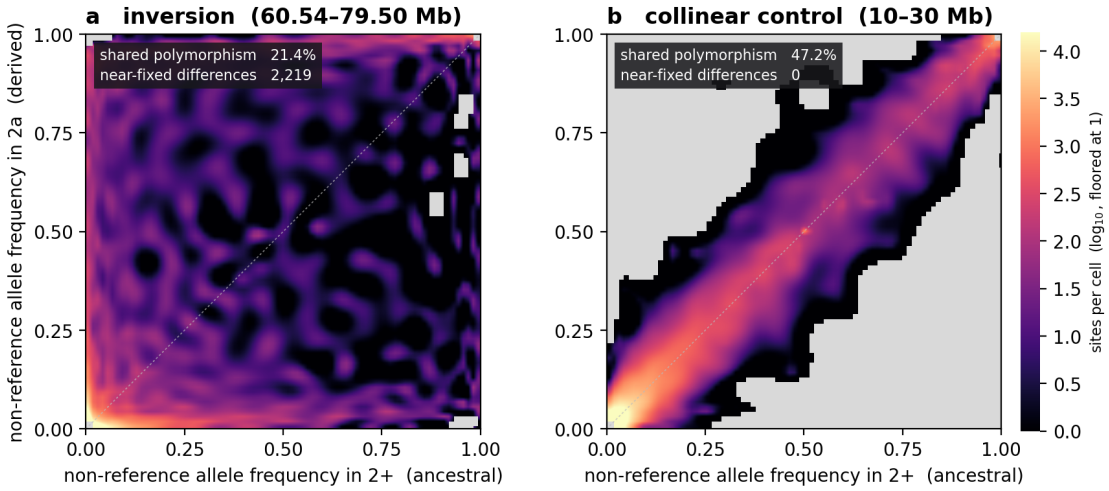


Figure S5: **Joint two-dimensional site-frequency spectrum** of non-reference allele frequency in the 2a (derived) versus 2+ (ancestral) arrangement. Each cell is a count of sites (colour, $\log_{10}$). In the inversion (a) density spreads off the diagonal and into the corners, with variants common in one arrangement and absent in the other, i.e. near-fixed differences (2,219 sites, 21% shared polymorphism), the direct signature of the recombination barrier. In the collinear control (b) the density collapses onto the diagonal (0 near-fixed differences, 47% shared), as expected for freely recombining, exchangeable sequence.

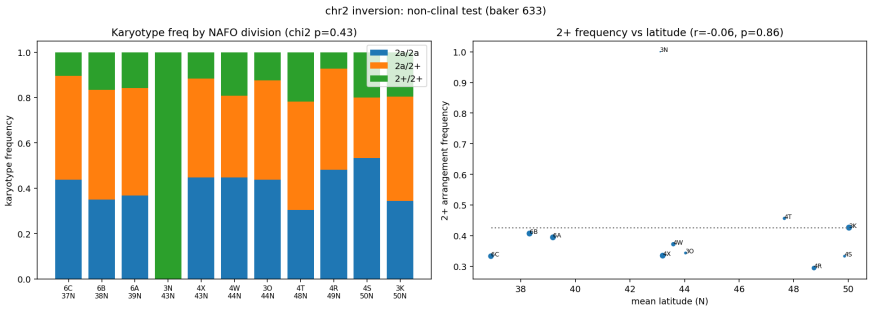


Figure S6: **The inversion is non-clinal.** Frequency of the ancestral 2+ arrangement across NAFO sampling divisions shows no latitudinal trend (see table S2). Geographic $F_{ST} \approx 0$, and the three karyotypes co-occur throughout the range.

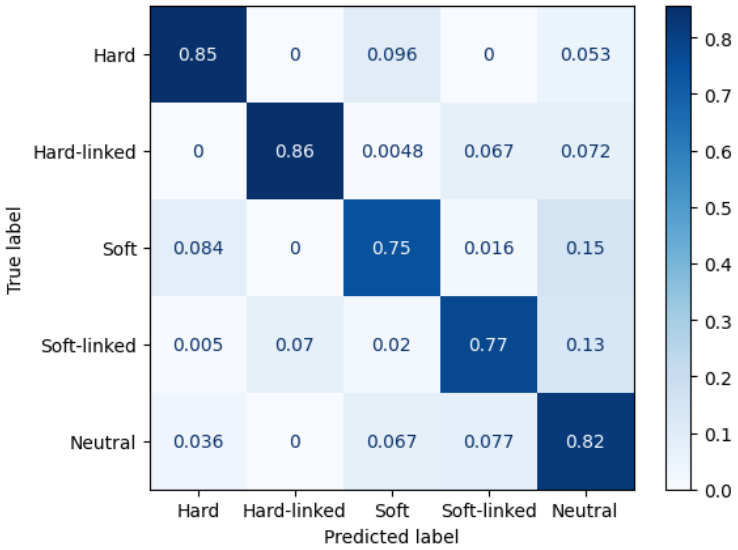


Figure S7: **Sweep-classifier confusion matrix** (five-class diploSHIC, full-SFS model, 81% overall accuracy). Recall is 0.85 for hard, 0.86 for linked-hard, 0.82 for neutral, 0.75 for soft and 0.77 for linked-soft sweeps.

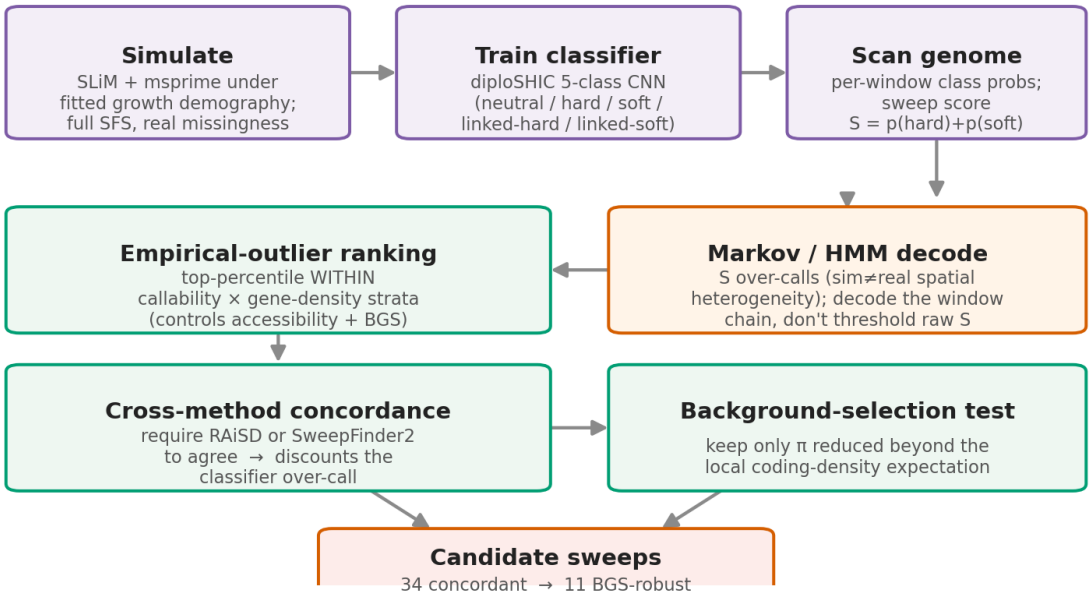


Figure S8: **The selective-sweep scan pipeline.** Simulate (SLiM + msprime under the fitted growth demography) → diploSHIC five-class classifier → Markov/HMM decode → stratified empirical-outlier ranking (within callability × gene-density strata) + cross-method concordance (RAiSD, SweepFinder2) + background-selection filter → candidate sweeps.

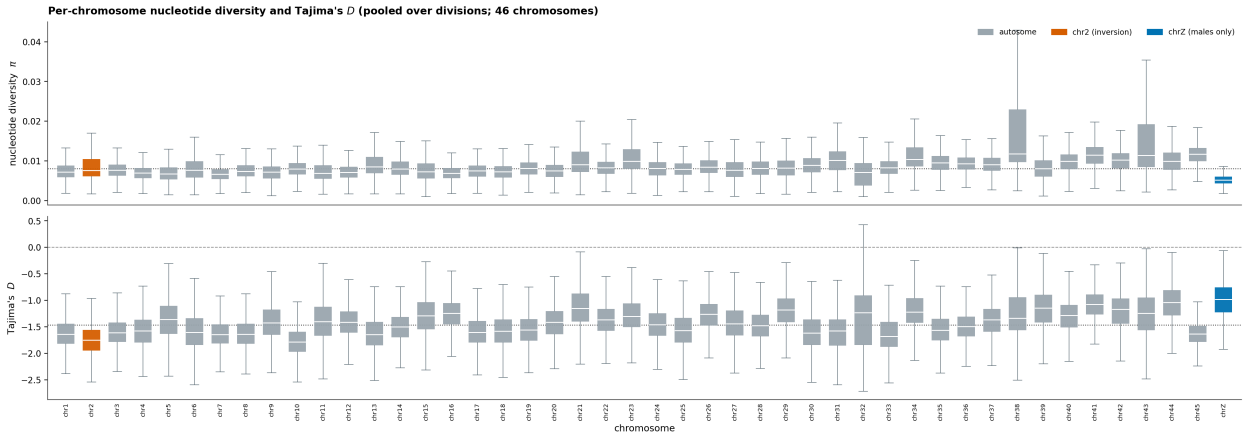


Figure S9: **Per-chromosome diversity.** Distribution of nucleotide diversity π (top) and Tajima's D (bottom) across 50 kb windows, one box per chromosome (pooled over the ten NAFO divisions, for all 46 chromosomes, the 45 autosomes plus chrZ). Diversity is uniform and D is strongly negative genome-wide, the signature of expansion. Two chromosomes stand apart. Chromosome 2 (the inversion, orange) carries an inflated upper tail of π and more-negative D from the segregating arrangements, and chrZ (blue, computed in the 330 hemizygous-informative males only, as females are Z-haploid) has reduced diversity (π median 0.005 vs ∼0.008 autosomal, ≈0.63×) as expected for a sex chromosome.

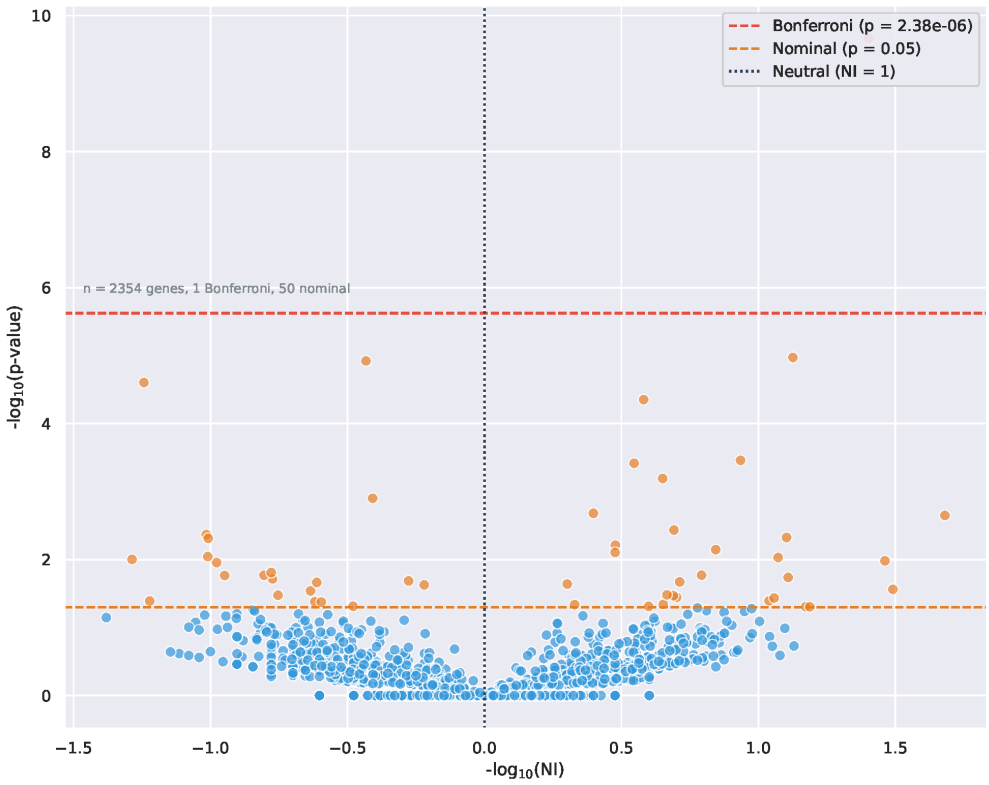


Figure S10: **Per-gene McDonald–Kreitman neutrality index** (mkado volcano plot, *I. coindetii* outgroup). Each point is one gene, plotted as $-\log_{10}$ of its neutrality index $\mathrm{NI} = (P_n/P_s)/(D_n/D_s)$ on the x-axis (right of the neutral $\mathrm{NI} = 1$ line = adaptive, $\mathrm{NI} < 1$, left = excess non-synonymous polymorphism, $\mathrm{NI} > 1$) against $-\log_{10} p$ on the y-axis. Dashed lines mark the nominal ($p = 0.05$) and Bonferroni thresholds. Of $n = 2{,}354$ testable genes, 50 are nominally significant and 1 survives Bonferroni. The genome-wide asymptotic $\alpha = 0.64$ resolves to few individual genes, as expected for a large-N_e species where per-gene MK is underpowered.

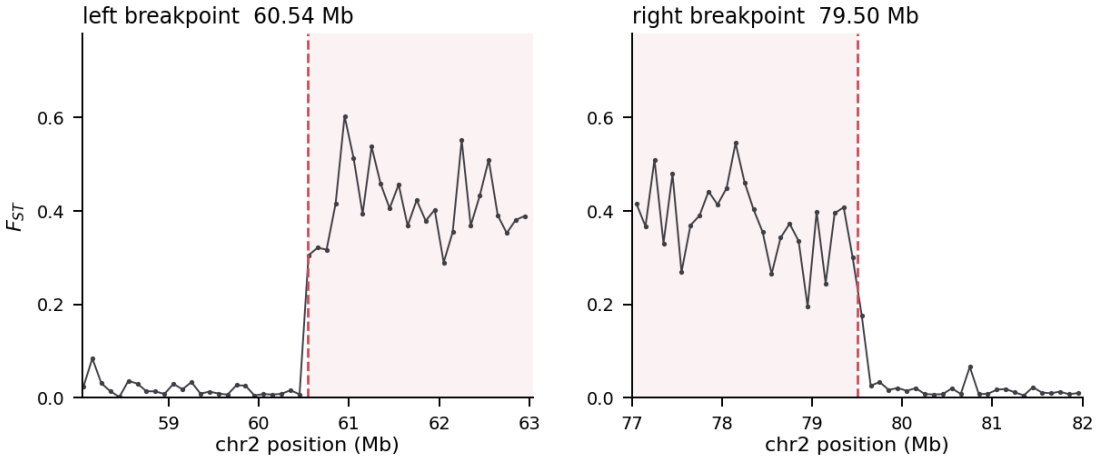


Figure S11: **Inversion breakpoints.** F_{ST} across the left (60.54 Mb) and right (79.50 Mb) breakpoints, a step rather than a ramp.

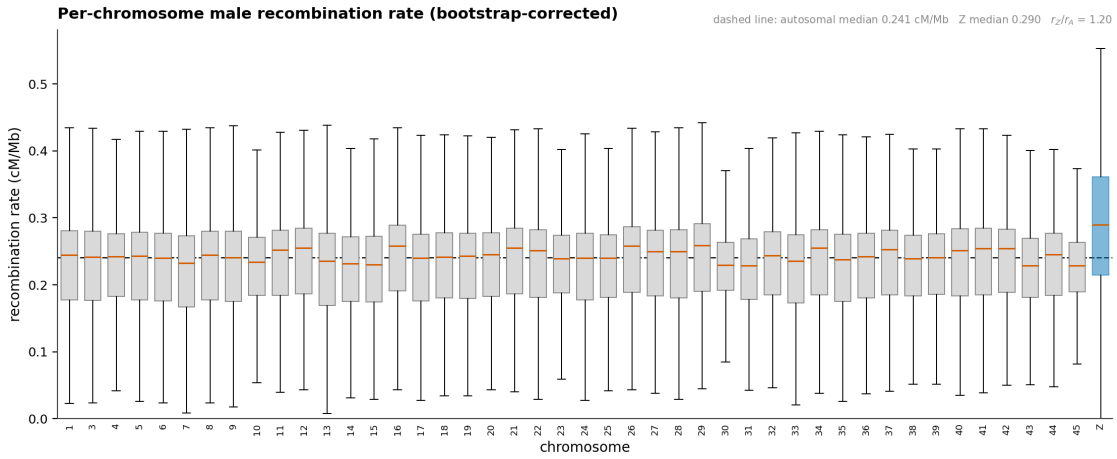


Figure S12: **Per-chromosome male recombination-rate distribution** (ReLERNN). Rates are corrected with 200 parametric-bootstrap replicates in each of 20 rate slices. The autosomes were predicted in 3–6-kb windows by one network, trained on chromosomes 1, 13 and 45 and applied to every autosome except chromosome 2. The Z was predicted in 30-kb windows by a separate network trained with the Z effective size set to 0.60 of the autosomal value (the Z-to-autosome diversity ratio), which places its rate on the autosomal scale. The dashed line marks the median of all autosomal windows (0.241 cM/Mb), and per-chromosome medians span 0.229–0.259 cM/Mb. The Z median is 0.290 cM/Mb, so $r_Z/r_A = 1.20$. Because the Z recombines only in ZZ males, equal recombination in the two sexes predicts $r_Z/r_A = 0.67$ and achiasmate female meiosis predicts 1.33. Chromosome 2 is not shown, because a window-scale map estimated from all karyotypes together cannot resolve a barrier that acts only between arrangements. That barrier is shown directly by the linkage-disequilibrium block spanning the inversion (Supplementary Fig. S4).

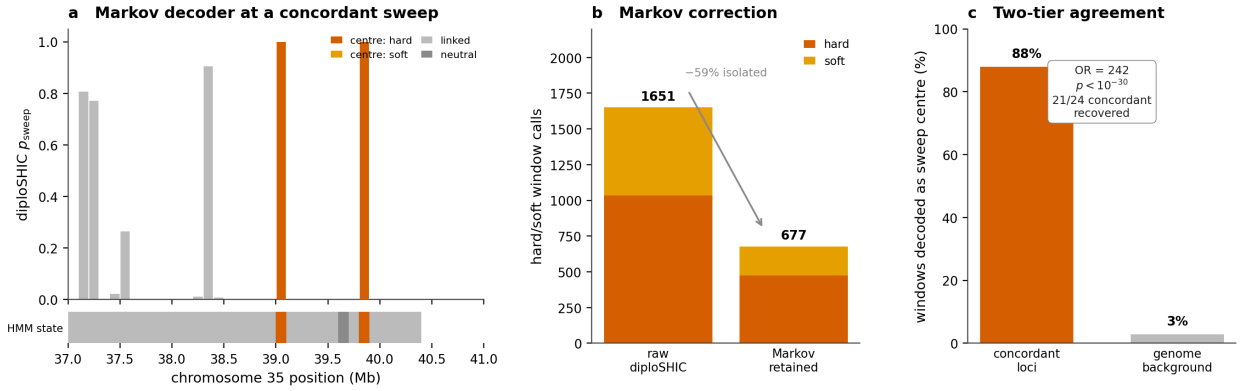


Figure S13: **Recombination-aware Markov decoder for the Tier-1 sweep calls.** A hidden Markov model over the per-100 kb-window diploSHIC classification (emissions = prior-corrected class posteriors, dwell lengths from the ∼400 kb sweep footprint scaled by the ReLERNN map, design-based rather than simulation-calibrated). **(a)** The decoder at a representative concordant sweep (chromosome 35, *ZEB2*), showing the sweep posterior $p_{\text{hard}}+p_{\text{soft}}$ per window and the Viterbi state track (neutral / linked / centre). It localises the centre inside the linked envelope and overrides spatially isolated high-posterior windows (e.g. near 38.35 Mb). **(b)** Markov correction reduces the raw hard/soft calls from 1,651 to 677 spatially coherent calls (−59%). **(c)** 88% of windows at the Tier-2 cross-method-concordant loci are decoded as sweep centres versus 2.9% genome-wide (odds ratio 242, $p < 10^{-30}$, 21 of 24 concordant regions recovered).

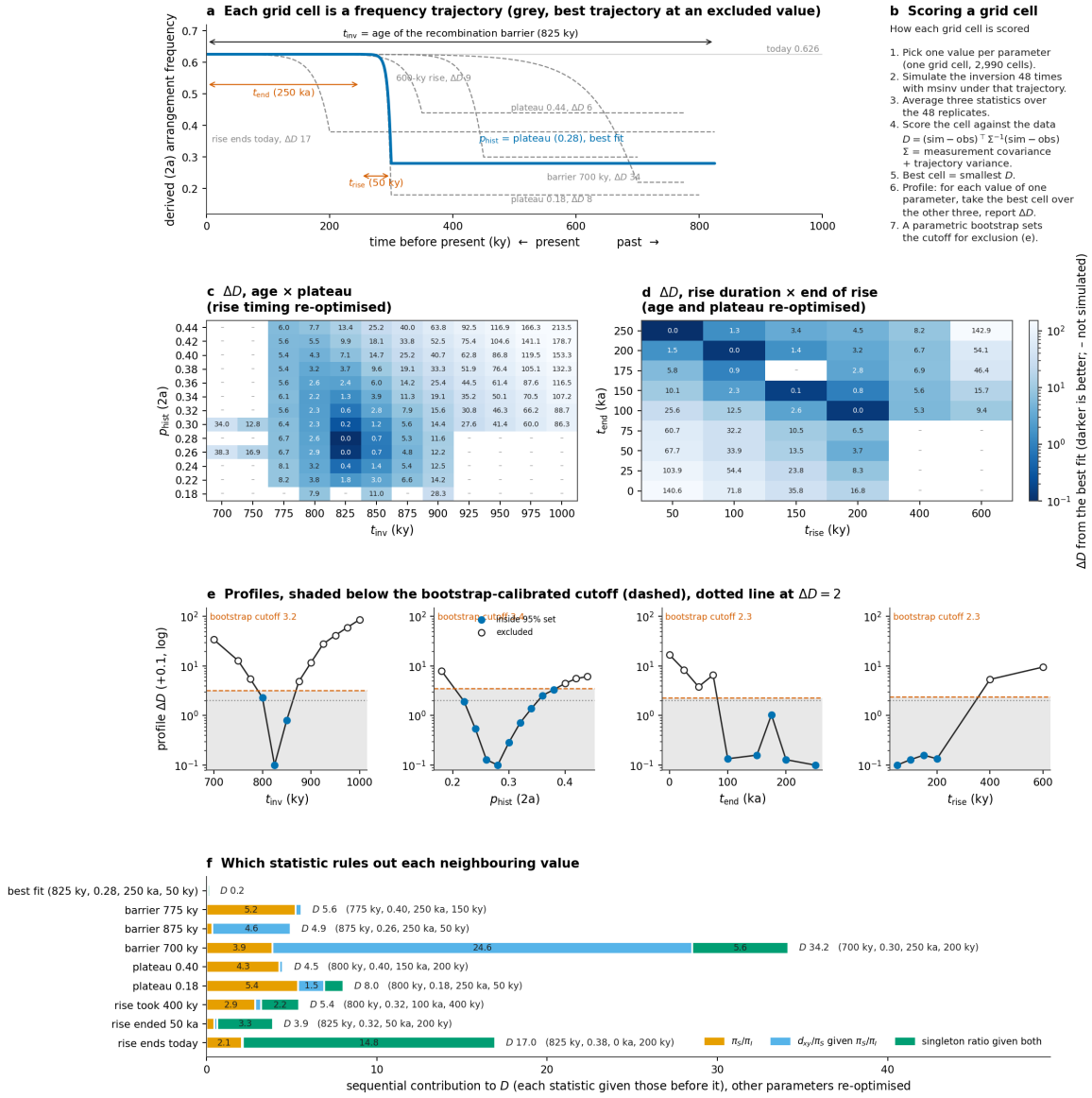


Figure S14: **Grid-search fit of the chromosome-2 inversion trajectory** (main-text Fig. 4). (a) Each grid cell is a deterministic frequency trajectory of the derived (2a) arrangement, set by the barrier age t_{inv}, the plateau frequency p_{hist}, the end of the rise t_{end} and its duration t_{rise}. The best fit is shown in blue, and each grey curve is the best trajectory at an excluded value of one parameter, labelled with its ΔD. (b) Each cell was simulated 48 times with msinv and scored by the distance D of π_S/π_I, d_{xy}/π_S and the singleton ratio from their empirical values, weighted by the measurement covariance plus the variance of the frequency trajectory under drift. (c, d) ΔD over barrier age and plateau, and over the duration and end of the rise, with the other two parameters re-optimised in each cell (dashes, not simulated). (e) Profile of each parameter with the other three re-optimised. The shaded band lies below the 95% cutoff calibrated by a parametric bootstrap (dashed line), and the dotted line marks $\Delta D = 2$. Filled points form the calibrated 95% set, a barrier age of 800–850 ky, a plateau of 0.22–0.38, and a rise of 50–200 ky that ended 100–250 ka. (f) Contribution of each statistic to D, taken in turn given the statistics before it, for the best fit and for the best cell at each excluded neighbouring value. The diversity and divergence ratios bound the age and the plateau, and the singleton ratio excludes a recent end to the rise.

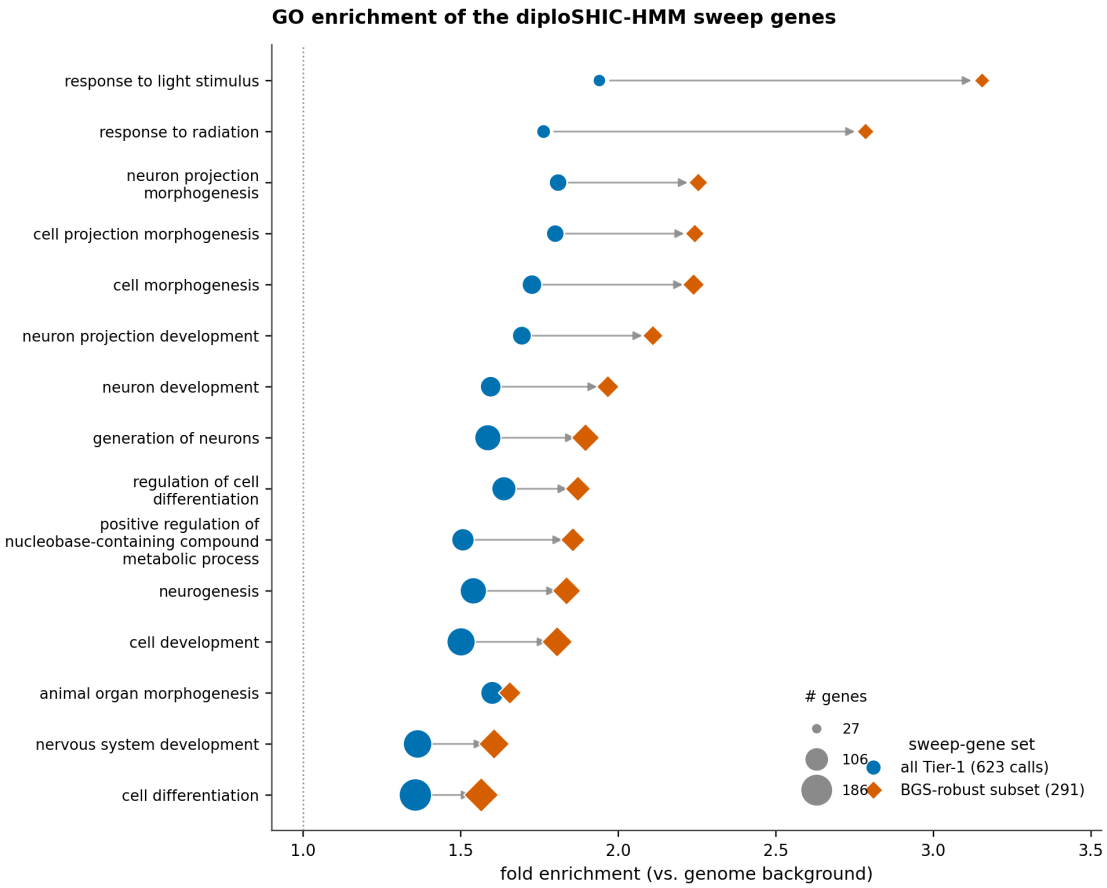


Figure S15: **GO enrichment of the diploSHIC-HMM sweep genes** (biological process, terms significant at FDR<0.05 in both sets). In this paired dot plot each term is shown for the full Tier-1 set (blue circle, 623 calls) and for its background-selection-robust subset (orange diamond, 291 calls), with x = fold enrichment and marker area = number of sweep genes carrying the term. The grey arrow marks the shift between the two sets. The BGS screen consistently sharpens the enrichment (rightward), and most strongly for the light-response / photoreception terms (response to light stimulus 1.9 $\rightarrow$ 3.1×, response to radiation 1.8 $\rightarrow$ 2.8×). Two coherent themes dominate, neural development and photoreception, and both survive the background-selection filter (table S4, full per-set enrichment tables are provided with the source). The hard-sweep calls are a sub-classification recorded in the calls table, not a separate gene set.

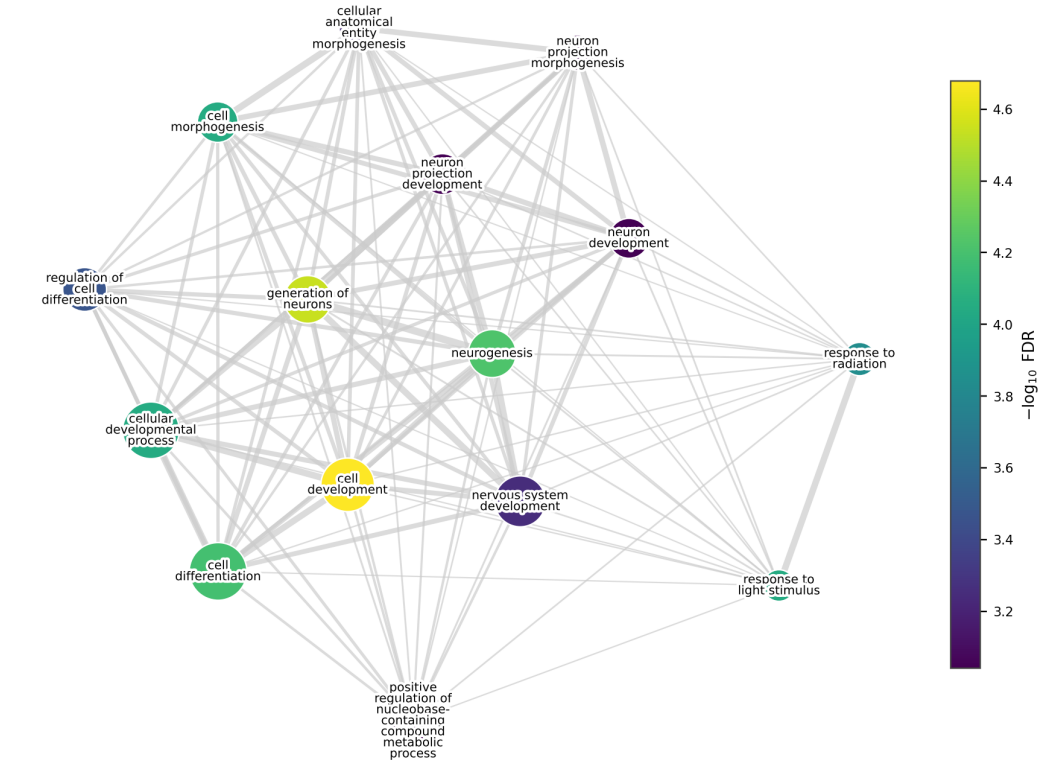


Figure S16: **GO enrichment map for the background-selection-robust sweep genes.** Built from the 291 BGS-robust Tier-1 calls (the subset whose local diversity reduction exceeds the background-selection expectation, 227 GO-annotated genes). Nodes are enriched biological-process terms (size = number of genes, colour = $-\log_{10}$ FDR). Edges connect terms sharing genes (Jaccard ≥ 0.15). The terms form connected modules of neural development / cell morphogenesis and photoreception (response to light/radiation), the same two themes seen in the GSEA dot plot (fig. S15).

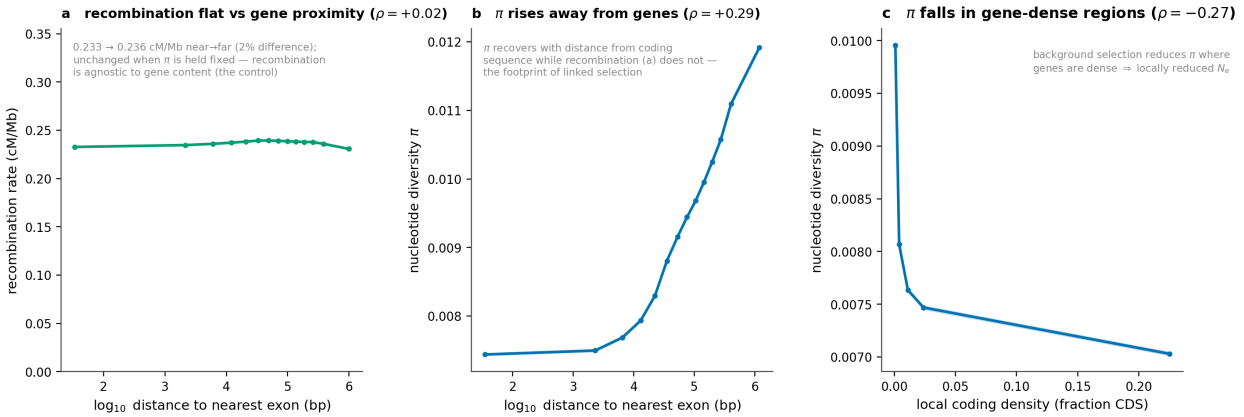


Figure S17: **Recombination, gene proximity and the efficacy of selection.** Per-window statistics (∼50 kb windows, autosomes). **(a)** As a control, recombination rate is essentially flat across distance from the nearest exon (0.233→0.236 cM/Mb near→far, a ∼2% difference, Spearman $\rho = +0.02$, unchanged when π is held fixed, partial $\rho = -0.01$), so recombination is agnostic to gene content, as it should be. **(b)** Nucleotide diversity π, by contrast, rises steeply with distance from coding sequence ($\rho = +0.29$, +43% near→far), the footprint of linked selection. **(c)** π falls with local coding density ($\rho = -0.27$), as background selection reduces diversity where genes are dense, i.e. a locally reduced effective size, and it is gene-density-driven. Linked selection in *Illex illecebrosus* is thus structured by gene density and gene proximity rather than by the recombination-rate axis, which is nearly uniform genome-wide (per-chromosome medians 0.23–0.26 cM/Mb, Supplementary Figs. S12, S22).

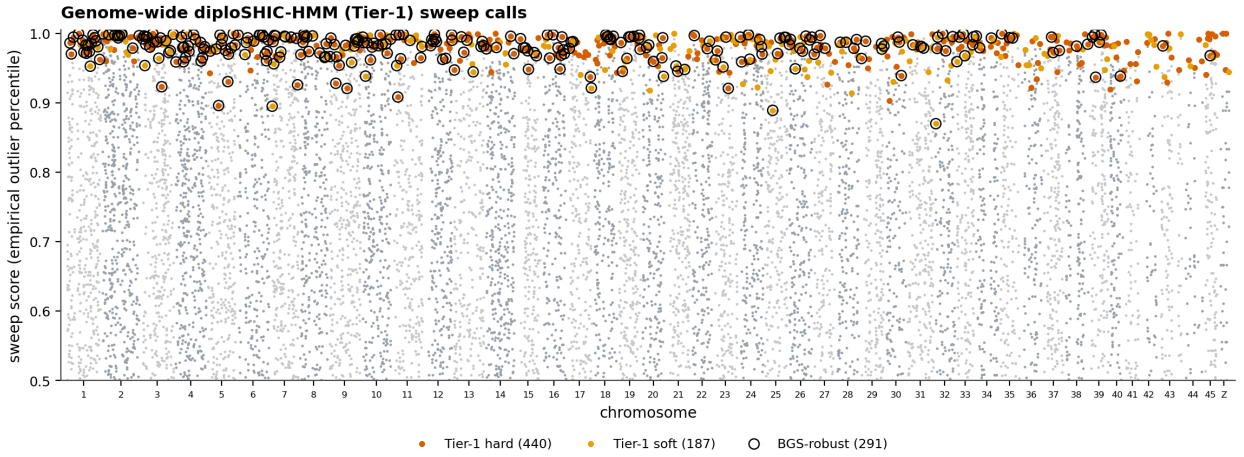


Figure S18: **Genome-wide diploSHIC-HMM (Tier-1) sweep calls.** Per-100 kb-window empirical-outlier sweep score (rank within callability × coding-density strata) across all 45 autosomes plus the Z chromosome (627 calls in total, 440 hard and 187 soft). The Markov-decoded Tier-1 calls are highlighted by class (hard, soft), and the background-selection-robust subset (§ BGS screen, 291 of the 623 autosomal calls) is ringed. The four chromosome-Z calls come from the male-only (n=330) diploSHIC scan and are diploSHIC-only (RAiSD/SweepFinder2 do not cover the Z). The calls are distributed across the genome with no single dominant locus.

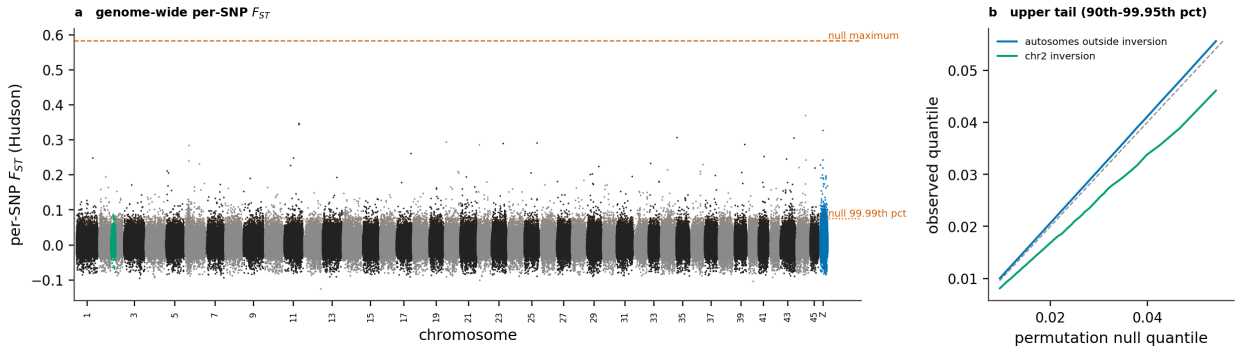


Figure S19: **Genome-wide per-SNP F_{ST} finds no differentiation outlier.** Per-SNP Hudson F_{ST} of fig. S2 along the genome (20,276,930 SNPs with minor-allele frequency above 0.05, every SNP shown). **(a)** No SNP exceeds the permutation-null maximum (0.583, dashed). The 2,327 SNPs above the null 99.99th percentile (0.075, dotted) on the autosomes match the number expected by chance and are spread across all chromosomes. The chromosome-2 inversion is green. chrZ (blue) exceeds the null tail (187 SNPs above the 99.99th percentile against 16 expected) because division sex ratios differ and females are hemizygous, and it is shown for completeness. **(b)** Upper-tail quantiles of the observed against the permutation null. The autosomes outside the inversion follow the identity line, and the inversion lies below it.

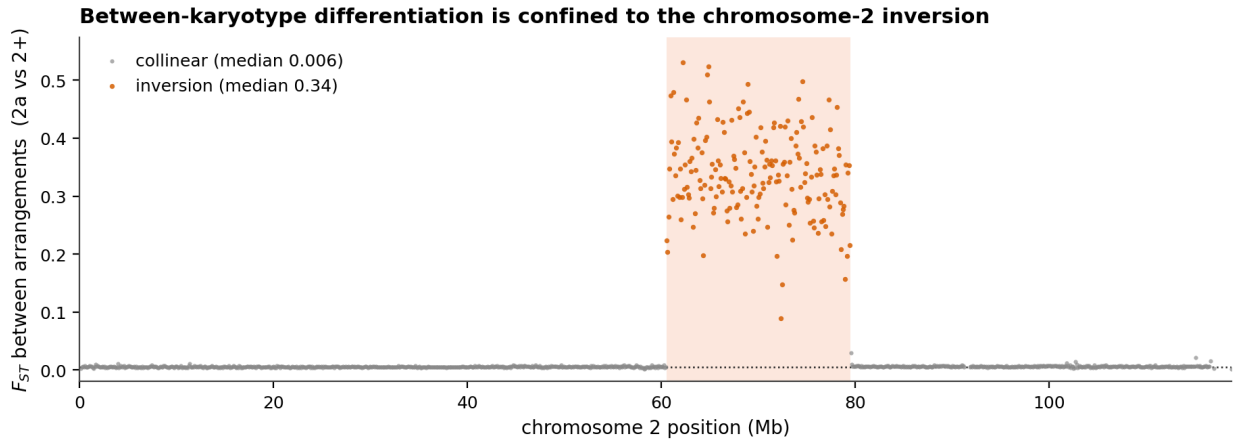


Figure S20: **Between-karyotype differentiation is confined to the chromosome-2 inversion.** Per-window F_{ST} between the 2a and 2+ arrangements (arrangement homozygotes, 2a/2a $n = 254$ and 2+/2+ $n = 95$) across the whole of chromosome 2, aggregated in 100 kb windows (ratio-of-averages Hudson estimator). Unlike the population-F_{ST} scan (which asks about geography), this asks how differentiated the two arrangements are along the chromosome. The collinear remainder (grey) sits at $F_{ST} \approx 0$ (median 0.006), so outside the inversion the karyotypes are one panmictic population, while the inversion (60.54–79.50 Mb, shaded) is strongly differentiated (median 0.34, max 0.53). The $F_{ST} \approx 0.37$ "structure" on chromosome 2 is thus between karyotypes and confined to the inversion, not geographic.

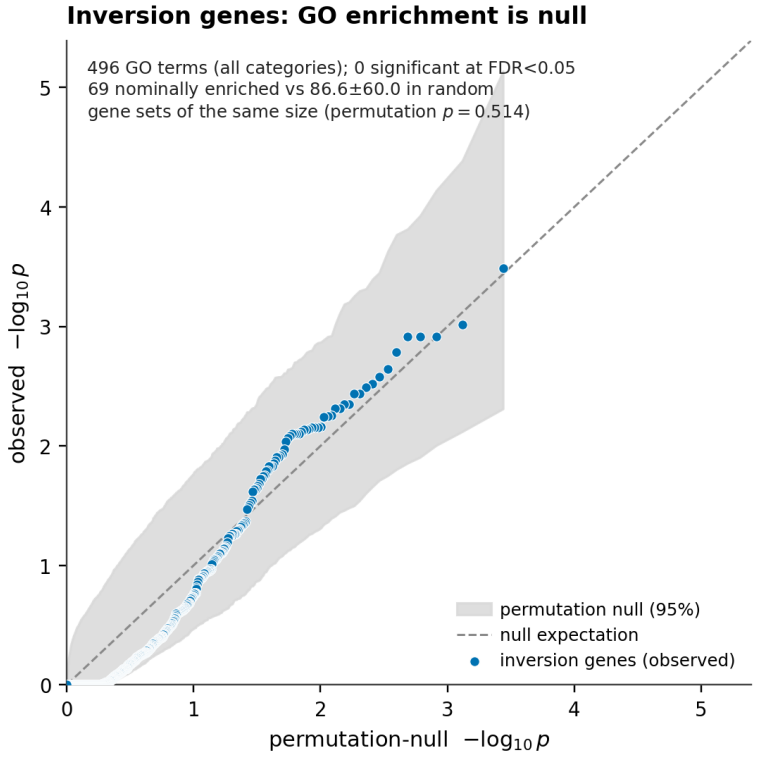


Figure S21: **The inversion-captured genes show no GO enrichment.** QQ-plot of the per-term GO-enrichment p-values (Fisher's exact vs. genome background, all three ontologies) for the genes in the inversion against a permutation null (500 random gene sets of the same size). A theoretical-uniform null is inappropriate for GO, because nested and correlated terms and tiny discrete counts inflate any small gene set, so the permutation band is the correct reference. The observed points fall entirely within it, and the inversion genes yield fewer nominally enriched terms than random sets of the same size (69 vs 86.6 ± 60, permutation $p = 0.51$) and none survive FDR<0.05. The large per-term fold values (up to $\sim$40$\times$) are small-count artefacts (most rest on 2 of the 29 GO-annotated genes), not enrichment (table S7).

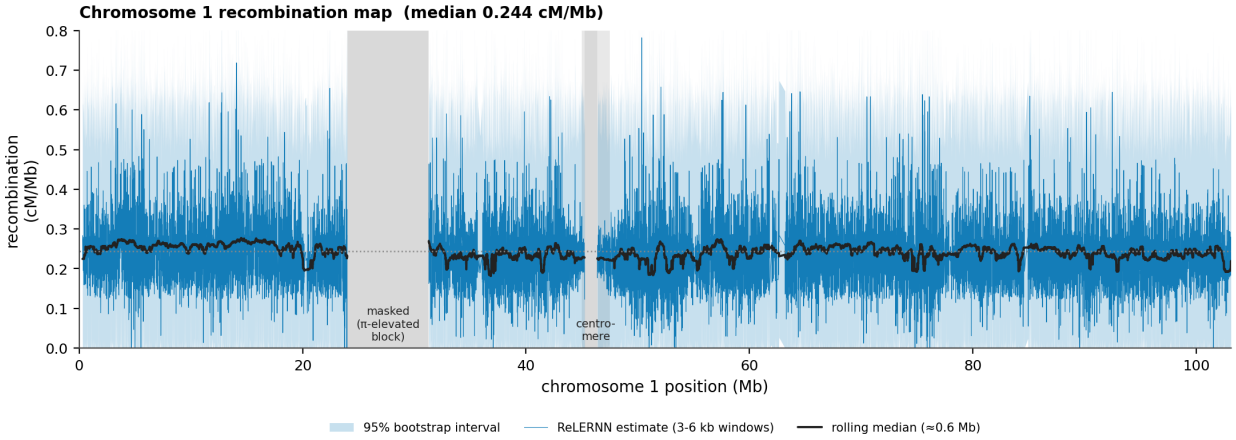


Figure S22: **The chromosome-1 recombination map, and what the LD-based map does not resolve.** Male ReLERNN estimate in 3–6-kb windows, corrected with 200 bootstrap replicates, with 95% bootstrap intervals and a 0.6-Mb rolling median (median 0.244 cM/Mb). Two stretches carry no windows. The π-elevated block at 24.0–31.3 Mb was masked. The 1.1-Mb gap at 45.25–46.39 Mb lies inside the 2.5-Mb block at 45.0–47.5 Mb, which is 80% repeat sequence (vs 52% chromosome-wide) and is the centromere. In the accessible pericentromeric flanks the estimate dips only $\sim$7%, although nucleotide diversity halves there (0.0038 vs 0.0084), and across the whole 103 Mb the 2-Mb rolling median spans just 0.22–0.27 cM/Mb ($-$10% to +10% of the median). The map's chromosome-scale dynamic range is therefore compressed, so centromeric suppression cannot be resolved and the nearly uniform landscape is in part an estimator ceiling.

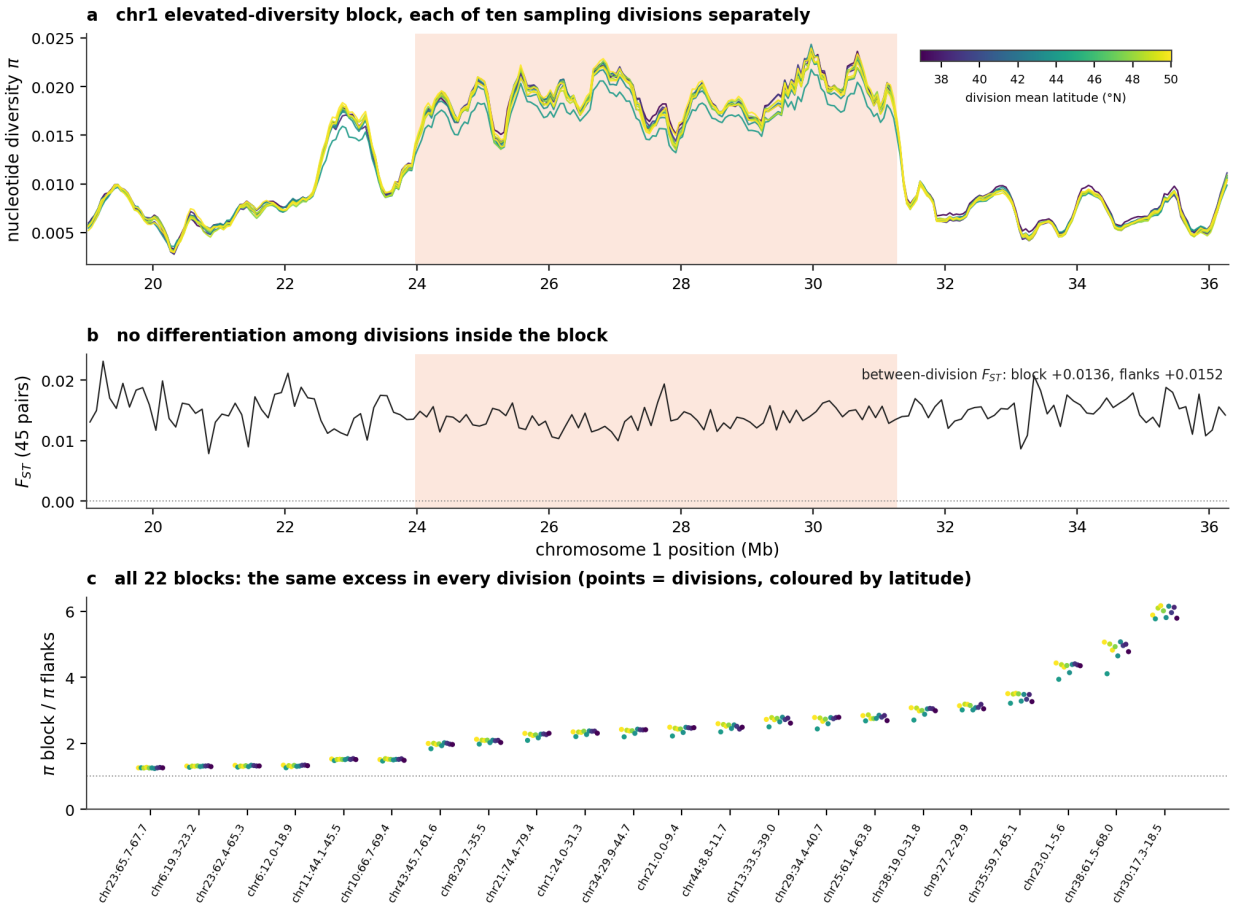


Figure S23: **The 22 elevated-diversity blocks are shared by every sampling division.** **(a)** Nucleotide diversity across chr1:19–36 Mb computed separately in each of the ten divisions with more than one individual (ANGSD, 50-kb windows, lines coloured by division latitude). The 24.0–31.3 Mb block (shaded) carries 2.2–2.4 times the flanking diversity in every division. **(b)** Between-division Hudson F_{ST} over the same span (all 45 division pairs, 100-kb windows) is lower inside the block than in the flanks (0.014 vs 0.015). The same holds for the three strongest blocks (chr30:17.3–18.5, chr23:0.1–5.6 and chr38:61.5–68.0 Mb, 0.006–0.007 inside vs 0.014–0.015 outside). **(c)** Block-to-flank diversity ratio per division for all 22 blocks. The excess varies by only 1–6% among divisions, so no block is driven by a subset of the sample. Read depth separates the blocks into collapsed duplications (depth raised in every individual, up to 2.2 times the flanks in the population sample and 2.7 and 2.4 times in long reads from the assembly individual and a second individual, with five to six times the flanking density of heterozygous sites at skewed allele fractions in single diploid genomes), under-mapped blocks (short-read depth 0.6–0.9 times the flanks, a third of the accessible sequence not uniquely mappable, and a much smaller deficit in long reads), and gene-poor regions of elevated diversity with normal depth (Supplementary Table S9). Genotype-likelihood diversity at mapping quality 60 retains most of the excess (for example chr30 6.4 to 5.8 times, chr23 5.2 to 4.0 times), because reads from unassembled copies map uniquely to the single assembled copy. All 22 blocks are masked from the sweep scan and gene-set analyses.

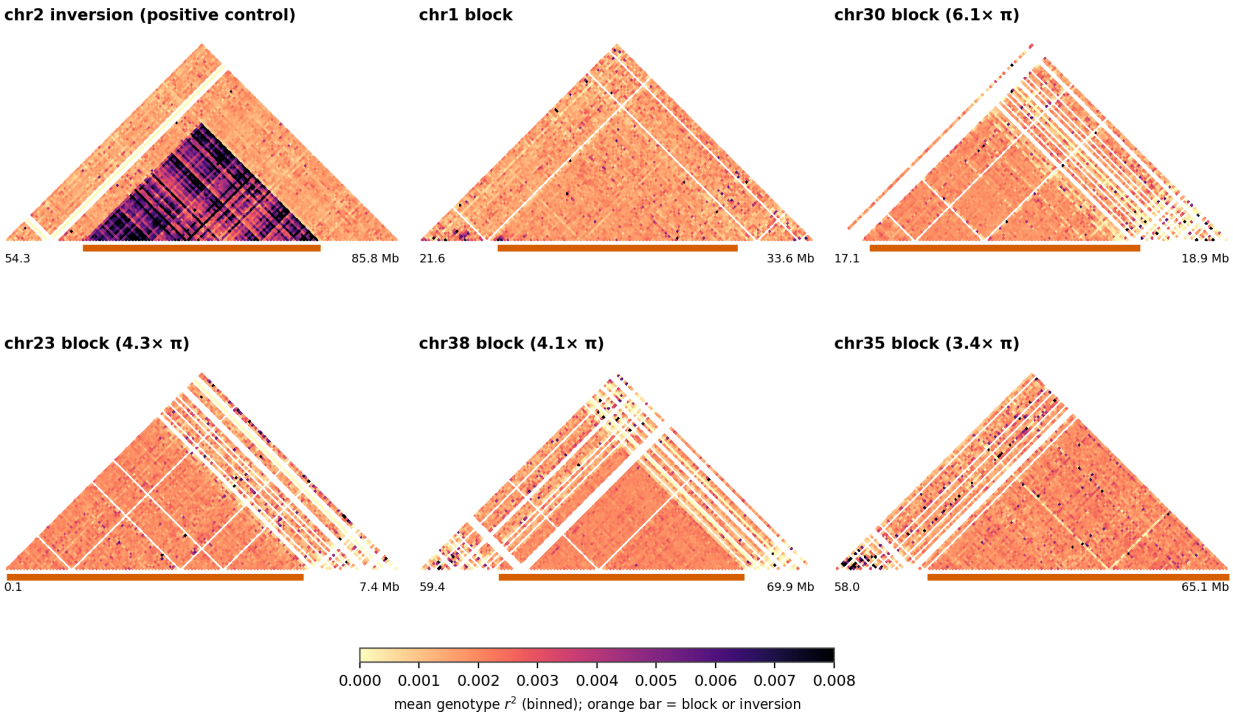


Figure S24: **The elevated-diversity blocks carry no inversion-like linkage disequilibrium.** Genotype r^2 among common SNPs (minor-allele frequency $\geq$0.1, $\leq$30% missing, 1,500 SNPs per region), averaged in 120 bins, for the chromosome-2 inversion (positive control) and five of the strongest blocks, each shown with a third of its length of flanking sequence. The orange bar marks the inversion or block. The inversion is a solid block of long-range LD over exactly its span (mean r^2 at >1 Mb 0.0056 inside vs 0.0016 in its flanks). None of the blocks differs from its flanks (0.0017–0.0020 vs 0.0012–0.0021 at >1 Mb, across all eight blocks tested), so none behaves as a segregating inversion or other long non-recombining haplotype. White stripes are bins with too few sampled SNPs.

Supplementary tables

Table S1: **Software and versions.** Versions were read off the analysis machine (conda metadata, `pip list`, `--version`, or recorded `versions.txt`). Citations are pointers to confirm at final formatting.

Tool (version)	Used for	Reference
msinv 0.3.5	chr2 inversion age/decline (inversion)	(Small, 2025; Kelleher et al., 2016)
grenepipe	read processing and variant calling	(Czech and Exposito-Alonso, 2022)
fastp	read trimming	(Chen et al., 2018)
BWA-MEM2	read mapping	(Vasimuddin et al., 2019)
Picard MarkDuplicates	duplicate removal	Broad Institute
GATK HaplotypeCaller	joint genotyping (gVCF, all sites)	(Poplin et al., 2018)
GenMap 1.3.0	mappability (150-mers, 2 mismatches)	(Pockrandt et al., 2020)
plink2	per-division allele counts (per-SNP Hudson F_{ST})	(Chang et al., 2015)
LiftOn	annotation transfer to *S. oualaniensis*	(Chao et al., 2024)
MACSE 2.07	codon alignment (sex-biased mutation)	(Ranwez et al., 2018)
EnTAP	functional annotation	(Hart et al., 2020)
ANGSD 0.940	SFS, θ, diversity (demography, diversity)	(Korneliussen et al., 2014)
Stairway Plot 2 (2.1.1)	demography	(Liu and Fu, 2020)
GONE2 1.0.4	recent demography	(Santiago et al., 2025)
moments 1.6.0	demographic inference (demography)	(Jouganous et al., 2017)
demes 0.2.3	demography interchange	(Gower et al., 2022)
ReLERNN 0.2	recombination map	(Adrion et al., 2020)
diploSHIC 1.1.0	sweep scan	(Kern and Schrider, 2018)
RAiSD (`0c705a8`)	sweep corroboration (sweep scan)	(Alachiotis and Pavlidis, 2018)
SLiM 5.2	forward simulation	(Haller et al., 2026)
msprime 1.4.2	coalescent sim, mutation overlay	(Baumdicker et al., 2022)
tskit 1.0.3	tree-sequence I/O	(Kelleher et al., 2018)
discoal 0.1.7a	sweep-sim validation	(Kern and Schrider, 2016)
pg_gpu 0.1.1.dev14	windowed $\pi/d_{xy}/F_{ST}$, PCA	(Pope et al., 2026)
mkado 0.5.0	McDonald–Kreitman	(Rivera-Colón et al., 2026)
DIAMOND 2.1.12	homology (inversion, sweep scan)	(Buchfink et al., 2021)
HMMER 3.4	Pfam domain screen	(Eddy, 2011)
Pfam-A	domain screen	(Mistry et al., 2021)
AnchorWave 1.2.3	*I. coindetii*↔*Illex illecebrosus* alignment	(Song et al., 2022)
minimap2 2.29	alignment	(Li, 2018)
degenotate 1.3	0-/4-fold degeneracy	(Thomas, 2024)
bedtools 2.30.0	interval operations	(Quinlan and Hall, 2010)
bcftools/samtools 1.21	VCF/BAM handling	(Danecek et al., 2021)
EarlGrey 7.0.1	TE annotation	(Baril et al., 2024)
StringTie 3.0.0	transcript assembly	(Kovaka et al., 2019)
SQANTI3	Iso-Seq QC	(Pardo-Palacios et al., 2024)

Table S2: **Chromosome-2 karyotype counts and derived-arrangement frequency by NAFO division** (ordered by latitude). The derived (2a) frequency p_{2a} shows no latitudinal cline (range 0.54–0.71, excluding 3N with one individual), consistent with geographic $F_{ST} \approx 0$ (fig. S6).

Division	2a/2a	2a/2+	2+/2+	N	p_{2a}	mean lat. (°N)
6C	38	40	9	87	0.667	36.9
6B	34	47	16	97	0.593	38.3
6A	35	45	15	95	0.605	39.2
3N	0	0	1	1	0.000	43.1
4X	42	41	11	94	0.665	43.2
4W	21	17	9	47	0.628	43.6
3O	7	7	2	16	0.656	44.0
4T	7	11	5	23	0.543	47.7
4R	27	25	4	56	0.705	48.7
4S	8	4	3	15	0.667	49.9
3K	35	47	20	102	0.574	50.0
Total	254	284	95	633	0.626	—

Table S3: **Use of the outgroups.** *I. coindetii* served three roles, the structural comparison of the chromosome-2 arrangements, a count of allele sharing at sites that differ between them, and the divergence term of the genome-wide McDonald–Kreitman (MK) test, the latter resting on a whole-genome *I. coindetii*↔*Illex illecebrosus* AnchorWave alignment. *I. argentinus* was aligned identically (CDS) and used as a secondary MK outgroup. It yields ∼9× fewer fixed coding differences than *I. coindetii* yet has genome-wide π indistinguishable from *Illex illecebrosus*, marking it as a recently-diverged congener of near-identical diversity. *S. oualaniensis* was used only for the structural comparison at the breakpoints, with positions given on the primary assembly and on HiC_scaffold_1.

Step	Quantity	Value
Alignment (AnchorWave v1.2.3, CDS-anchored)		
	I. coindetii substitution sites (genome-wide)	45,722,335
	aligned scaffold identity (inversion)	99.2–99.9%
Chromosome-2 arrangements in the reference individual (minimap2)		
	hap1 vs. primary, inverted segment	60.54–79.53 Mb
	hap2 vs. primary	collinear
	arrangement of hap1 / hap2	2a / 2+
	I. coindetii 37.95–74.95 Mb, aligned to 2+	one inverted run
	… aligned to 2a	two runs, split at 60.5 Mb
	I. coindetii adjacency at 56.9 Mb	kept in 2+, broken in 2a
Chromosome-2 breakpoints in *S. oualaniensis* (HiC_scaffold_1, minimap2)		
	Illex illecebrosus chr2 genes transferred to HiC_scaffold_1	265 of 319
	left breakpoint, genes at 60.49 and 60.55 Mb	adjacent (51.52, 51.57 Mb)
	right breakpoint, 78.2–83.6 Mb	one inverted run
	2+ adjacencies (hap2), both breakpoints	present
	2a adjacencies (hap1), both breakpoints	absent
Allele sharing with *I. coindetii* (inversion, frequency difference between 2a and 2+ ≥ 0.7)		
	differentiated sites (*I. coindetii* base callable)	13,998 (13,582)
	… 2a majority allele = *I. coindetii*	25.0%
	I. coindetii differs from the reference base	4,175
	… 2a / 2+ carries the *I. coindetii* allele	63.0% / 37.0%
McDonald–Kreitman test (mkado, *I. coindetii* divergence)		
	genes tested	20,999
	nonsynonymous / synonymous divergence (Dn / Ds)	15,343 / 13,670
	nonsynonymous / synonymous polymorphism (Pn / Ps)	468,124 / 630,949
	asymptotic α (95% CI)	0.642 (0.539–3.673)
I. argentinus (secondary MK outgroup, CDS alignment)		
	nonsynonymous / synonymous divergence (Dn / Ds)	1,355 / 1,805
	fixed coding differences (vs *I. coindetii* 29,013)	3,160 (∼9× fewer)
	asymptotic α	0.526
	genome-wide π (ANGSD, 6 autosomes; *Illex illecebrosus* = 0.0093)	0.0092

Table S4: **Gene-ontology enrichment of the Tier-1 (diploSHIC-only, Markov-corrected) sweep genes.** The 623 Tier-1 calls overlap 885 genes (475 GO-annotated). Enrichment by Fisher's exact test against the annotated-gene background on the scanned chromosomes, Benjamini–Hochberg FDR. 124 biological-process terms reach FDR<0.05, and representative terms are shown, spanning the two dominant themes (neural development and RNA-regulatory or transcriptional control). All are carried by the hard-sweep calls. The soft-sweep gene set has no term at FDR<0.05. The full machine-readable table is `hmm_decode_masked/tier1_GO_enrichment.tsv`.

GO id	term (biological process)	genes (T1/bg)	fold	FDR
GO:0048699	generation of neurons	123/1441	1.59	2.0×10^{-4}
GO:0022008	neurogenesis	125/1508	1.54	2.6×10^{-4}
GO:0030182	neuron differentiation	91/1075	1.57	1.6×10^{-3}
GO:0030154	cell differentiation	186/2548	1.36	5.4×10^{-4}
GO:0009887	animal organ morphogenesis	96/1115	1.60	8.9×10^{-4}
GO:0050767	regulation of neurogenesis	66/693	1.77	1.3×10^{-3}
GO:0000902	cell morphogenesis	72/775	1.73	1.3×10^{-3}
GO:0007409	axonogenesis	45/463	1.81	1.0×10^{-2}
GO:0048858	cell projection morphogenesis	59/609	1.80	1.6×10^{-3}
GO:0051252	regulation of RNA metabolic process	145/1920	1.40	1.3×10^{-3}
GO:2001141	regulation of RNA biosynthetic process	135/1768	1.42	1.6×10^{-3}

Table S5: **Functional clusters of the background-selection-robust Tier-1 sweep genes.** The 151 enriched biological-process GO terms (FDR<0.05) of the BGS-robust set (fig. S15), grouped into functional clusters by term semantics. Neural development and cell morphogenesis/differentiation dominate, with RNA-regulatory control and organ development, and a distinct photoreception/sensory cluster.

functional cluster	terms	fold range	representative term (min FDR)
Neural development	21	1.6–22.2	generation of neurons (3×10^{-5})
Cell morphogenesis & differentiation	17	1.5–2.2	cell development (2×10^{-5})
RNA / transcription regulation	14	1.5–2.1	reg. RNA metabolic process (9×10^{-4})
Organ / anatomical development	13	1.2–2.5	system development (3×10^{-3})
Photoreception / sensory response	7	1.2–5.8	response to light stimulus (1×10^{-4})
Other regulation / metabolism	65	1.2–22.2	reg. developmental process

Table S6: **Tier-2 (cross-method-concordant) sweep genes.** The 24 genome-wide-scan regions where the diploSHIC-HMM sweep signal is independently corroborated by an LD/SFS-footprint method (RAiSD or SweepFinder2). Nine regions carry a named protein-coding gene (14 genes, listed). The remaining concordant regions are gene deserts or uncharacterised loci. Coordinates are in Mb, and α and DoS are from the McDonald–Kreitman test against *I. coindetii* ('–' where the test is unpowered, i.e. too few fixed/polymorphic sites). The full candidate set, comprising all 227 diploSHIC-outlier regions, the 24 concordant regions and the per-region background-selection test, is provided as machine-readable tables (`outlier_scan_45_masked/{windows,regions}.tsv` and the `concordant` gene/MK/BGS tables).

region (Mb)	sweep	gene	EnTAP best hit	α	DoS
1:20.3	hard	*Aplnr*	Apelin receptor	1.00	+0.17
1:85.5	hard	*eif3h*	Eukaryotic translation initiation factor 3	0.69	+0.20
1:85.5	hard	*babam1*	BRISC and BRCA1-A complex member 1	–	-0.48
1:85.5	hard	*ttc39b*	Tetratricopeptide repeat protein 39B	–	-0.27
2:84.8	hard	*NDRG3*	Protein NDRG3	–	-0.18
5:38.0	hard	*cept1*	Choline/ethanolaminephosphotransferase 1	–	-1.00
5:38.0	hard	*Arsi*	Arylsulfatase I	–	-0.25
6:3.0	hard	*Rexo1*	RNA exonuclease 1 homolog	–	-0.53
13:73.2	hard	*Rab3*	Ras-related protein Rab-3	–	-0.17
13:73.2	hard	*hyi*	Putative hydroxypyruvate isomerase	–	-0.64
13:73.2	hard	*Vps28*	Vacuolar protein sorting-associated protein 28	–	–
32:9.3	soft	*Macrod2*	ADP-ribose glycohydrolase MACROD2	–	-0.59
33:9.0	hard	*Tinagl1*	Tubulointerstitial nephritis antigen-like	–	-0.23
35:39.0	hard	*zag-1*	Zinc finger E-box-binding homeobox (ZEB)	–	-0.31

Table S7: **Genes captured by the chromosome-2 inversion** (chr2:60.54–79.50 Mb). The 45 functionally annotated genes (of 89 gene models), ordered by position; *fixed* = number of exonic fixed differences between the 2a and 2+ arrangements ($|\Delta AF| \geq 0.9$; '–' none). EnTAP best-hit descriptions. The set spans lipid metabolism (phospholipases, PNPLA, malonyl-/acyl-CoA transferases), peroxisomal and mitochondrial function, DNA repair (XRCC1, TDG, WRNIP1), transcription/splicing (ERG, PRPF8, WDR82) and signalling (PRKCD), with no coherent pathway — consistent with the absent GO enrichment (fig. S21). Gene IDs are the `LOC_` suffix.

region (Mb)	gene	EnTAP best hit	fixed
2:60.6	00005299	protein HGH1 homolog isoform X1	–
2:60.9	00005300	Phospholipase A2 domain-containing protein	2
2:61.4	00005301	lon protease homolog 2, peroxisomal isoform X2	–
2:61.4	00005302	39S ribosomal protein L18, mitochondrial	–
2:61.4	00005303	sodium- and chloride-dependent taurine transporter	–
2:61.6	00005304	cytochrome P450 4X1-like	4
2:61.6	00005305	uncharacterized protein LOC115219520	–
2:64.1	00005315	PNPLA domain-containing protein	–
2:64.1	00005316	PNPLA domain-containing protein	–
2:65.5	00005319	CCHC-type domain-containing protein	–
2:65.7	00005320	uncharacterized protein LOC127831186	–
2:66.1	00005322	solute carrier family 66 member 2 isoform X2	–
2:66.3	00005324	Ubiquitin-conjugating enzyme E2 T	–
2:68.0	00005330	PCDHD1 (protocadherin)	–
2:68.8	00005334	retrovirus-related Pol polyprotein (transposon)	–
2:69.0	00005337	PNPLA domain-containing protein	–
2:69.0	00005338	ELMO domain-containing protein	–
2:69.1	00005339	malonyl-CoA-acyl carrier protein transacylase, mito.	1
2:69.1	00005340	acidic repeat-containing protein-like	4
2:69.1	00005341	RPGRIP1L (ciliary)	2
2:70.2	00005347	transcriptional regulator ERG homolog	–
2:71.2	00005349	transcriptional regulator ERG homolog	–
2:71.6	00005350	uncharacterized protein LOC115215517	1
2:71.9	00005351	cysteine-rich venom protein	2
2:72.0	00005352	cysteine-rich venom protein	–
2:74.6	00005356	DUF7043 domain-containing protein	–
2:77.7	00005360	SWIM-type domain-containing protein	–
2:77.8	00005361	UPB1 (β-ureidopropionase)	–
2:77.8	00005362	DNA-repair protein XRCC1 (N-terminal domain)	–
2:77.8	00005363	TIM21 (mitochondrial import)	–
2:77.9	00005364	dynein light chain roadblock-type 2	–
2:77.9	00005365	TDG (thymine-DNA glycosylase)	–
2:78.0	00005367	small integral membrane protein 4	–
2:78.1	00005368	uncharacterized protein LOC115226124	–
2:78.1	00005369	ATPase WRNIP1-like	–
2:78.2	00005371	unnamed protein product (*Sepia*)	1
2:78.2	00005372	PPM1H (protein phosphatase)	–
2:78.2	00005373	WDR82	–
2:78.2	00005374	glycerate kinase-like	–
2:78.4	00005375	PRKCD (protein kinase C δ)	–
2:78.6	00005376	acyl-coenzyme A amino acid N-acyltransferase 1-like	–
2:78.7	00005378	uncharacterized protein LOC115226077	–
2:78.7	00005379	PRPF8 (pre-mRNA splicing factor)	–

region (Mb)	gene	EnTAP best hit	fixed
2:78.8	00005380	synaptic vesicle 2-related protein-like	–
2:79.3	00005384	unnamed protein product (*Sepia*)	–

Table S8: **Sampling events.** Every collection event (NAFO division, date, depth) with its source, the number of individuals analysed in this study (all 633 whole-genome sequenced individuals) and the number retained by Baker et al. (2025) after their genotype-missingness filter (540). Divisions are ordered north to south by mean sampling latitude. 4X was bulk-sampled across the division with no per-tow depth or position recorded.

Division	Date	Source	Depth (m)	N (this study)	N (Baker)
3K	16 August 2023	Recreational jig	3	48	48
	30 August 2023	Coastal fishery jig	5	54	46
4S	28 August 2023	Canadian trawl survey	228	2	2
	29 August 2023	Canadian trawl survey	251	1	1
	30 August 2023	Canadian trawl survey	80	1	1
	30 August 2023	Canadian trawl survey	124	2	2
	31 August 2023	Canadian trawl survey	152	2	1
	31 August 2023	Canadian trawl survey	155	3	3
	1 September 2023	Canadian trawl survey	273	1	1
	1 September 2023	Canadian trawl survey	296	1	1
	3 September 2023	Canadian trawl survey	243	2	2
4R	7 August 2023	Canadian trawl survey	113	36	24
	25 August 2023	Canadian trawl survey	173	2	2
	25 August 2023	Canadian trawl survey	233	5	4
	25 August 2023	Canadian trawl survey	245	9	7
	28 August 2023	Canadian trawl survey	221	4	3
4T	2 September 2023	Canadian trawl survey	243	1	0
	7 September 2023	Canadian trawl survey	27	1	1
	8 September 2023	Canadian trawl survey	49	1	0
	9 September 2023	Canadian trawl survey	112	1	0
	9 September 2023	Canadian trawl survey	183	1	0
	9 September 2023	Canadian trawl survey	306	1	0
	10 September 2023	Canadian trawl survey	81	1	1
	10 September 2023	Canadian trawl survey	101	1	1
	10 September 2023	Canadian trawl survey	285	1	0
	12 September 2023	Canadian trawl survey	285	1	0
	15 September 2023	Canadian trawl survey	50	1	1
	22 September 2023	Canadian trawl survey	39	1	1
	22 September 2023	Canadian trawl survey	142	2	1
	22 September 2023	Canadian trawl survey	158	1	1
	23 September 2023	Canadian trawl survey	341	2	2
	24 September 2023	Canadian trawl survey	118	1	0

Division	Date	Source	Depth (m)	N (this study)	N (Baker)
	27 September 2023	Canadian trawl survey	36	1	0
	28 September 2023	Canadian trawl survey	56	1	1
	28 September 2023	Canadian trawl survey	62	1	1
	29 September 2023	Canadian trawl survey	49	2	1
3O	13 October 2023	Canadian trawl survey	204	14	7
	14 October 2023	Canadian trawl survey	350	1	1
	14 October 2023	Canadian trawl survey	613	1	1
4W	16 July 2023	Canadian trawl survey	84	47	43
4X	July 2022	Canadian trawl survey	not recorded	94	76
3N	20 October 2023	Canadian trawl survey	160	1	1
6A	22 September 2022	US trawl survey	120	95	84
6B	18 June 2023	Commercial trawl	74	49	46
	16 August 2023	Commercial trawl	201	48	48
6C	14 September 2022	US trawl survey	307	87	73
Total				633	540

Table S9: **Anomalous-diversity blocks, masked from the sweep scan and gene-set analyses.** Blocks of ≥1 Mb whose rolling nucleotide diversity exceeds twice the chromosome median. π fold is the block mean over the chromosome median. Division ratio is the range of block-to-flank diversity across the ten sampling divisions (ANGSD). Depth is block-to-flank read depth, as the median over all 633 individuals (Illumina, mapping quality ≥0) and for the assembly individual (PacBio HiFi, mapping quality ≥20). Non-unique is the percentage of the block's accessible sequence that 150-bp reads cannot map uniquely (GenMap, up to two mismatches), against 3.8% genome-wide. Class is collapsed duplication (population depth ≥1.15), under-mapped (depth ≤0.90), or elevated diversity with normal depth. In long-read genotypes of two individuals, collapsed blocks carry five to six times the flanking density of heterozygous sites at skewed allele fractions (median minor-allele fraction 0.29–0.33 vs 0.38–0.42 in flanks), and elevated-diversity blocks carry about twice the flanking density at normal fractions (chr1, 0.41 vs 0.40–0.43). The under-mapped chr21:0.0–9.4 block also shows skewed fractions (0.25 and 0.28 vs 0.42 and 0.44), with heterozygous-site density halved in the assembly individual, whose second haplotype largely fails to map, and tripled in the second individual, consistent with under-mapped blocks being largely duplicated within the assembly (33% of their accessible sequence non-unique).

Block	Mb	π fold	D block / chrom	Genes/Mb	Division ratio	Depth pop. / HiFi	Non-unique (%)	Class
chr1:24.0–31.3	7.3	2.8	−1.77 / −2.21	1.8	2.20–2.37	0.97 / 1.05	7	elevated diversity
chr6:12.0–18.9	6.9	2.3	−1.58 / −2.17	1.7	1.26–1.34	0.62 / 0.92	45	under-mapped
chr6:19.3–23.2	3.9	2.4	−1.56 / −2.17	1.8	1.27–1.32	0.66 / 0.84	41	under-mapped
chr8:29.7–35.5	5.8	2.3	−1.97 / −2.22	1.7	1.97–2.12	0.98 / 1.10	7	elevated diversity
chr9:27.2–29.9	2.7	3.6	−1.12 / −2.12	2.2	3.01–3.19	1.19 / 1.18	9	collapsed duplication
chr10:66.7–69.4	2.7	2.1	−2.06 / −2.26	0.4	1.46–1.53	0.95 / 1.01	2	elevated diversity
chr11:44.1–45.5	1.4	2.1	−1.83 / −2.15	2.1	1.48–1.53	0.97 / 1.04	3	elevated diversity
chr13:33.5–39.0	5.5	2.7	−1.79 / −2.19	3.1	2.50–2.79	0.93 / 1.05	4	elevated diversity
chr21:0.0–9.4	9.4	2.6	−1.39 / −1.94	1.7	2.22–2.49	0.68 / 0.61	36	under-mapped
chr21:74.4–79.4	5.0	2.5	−1.42 / −1.94	2.0	2.09–2.30	0.66 / 0.61	39	under-mapped
chr23:0.1–5.6	5.5	4.3	−0.96 / −2.04	2.2	3.94–4.43	1.78 / 1.91	12	collapsed duplication
chr23:62.4–65.3	2.9	2.3	−1.68 / −2.04	2.8	1.28–1.33	0.88 / 1.11	8	under-mapped
chr23:65.7–67.7	2.0	2.3	−1.73 / −2.04	1.5	1.23–1.27	1.07 / 1.12	10	elevated diversity
chr25:61.4–63.8	2.4	2.8	−1.57 / −2.17	2.9	2.68–2.86	1.16 / 1.16	25	collapsed duplication
chr29:34.4–40.7	6.3	2.6	−1.41 / −1.97	2.1	2.44–2.79	0.86 / 0.98	30	under-mapped
chr30:17.3–18.5	1.2	6.1	−0.69 / −2.17	2.5	5.77–6.17	2.19 / 2.66	2	collapsed duplication
chr34:29.9–44.7	14.8	2.6	−1.45 / −1.95	1.8	2.20–2.43	0.81 / 0.93	33	under-mapped
chr35:59.7–65.1	5.4	3.4	−1.47 / −2.13	1.9	3.21–3.52	1.53 / 1.84	4	collapsed duplication
chr38:19.0–31.8	12.8	2.8	−1.41 / −1.97	2.2	2.70–3.08	0.87 / 1.17	30	under-mapped
chr38:61.5–68.0	6.5	4.1	−1.08 / −1.97	2.6	4.11–5.08	1.44 / 1.21	17	collapsed duplication
chr43:45.7–61.6	15.9	2.4	−1.52 / −1.84	2.2	1.83–2.02	0.93 / 0.94	31	elevated diversity
chr44:8.8–11.7	2.9	2.7	−1.27 / −1.85	1.4	2.35–2.59	0.96 / 0.88	5	elevated diversity

Table S10: **Z recombination and sex-biased mutation.** Top, the ratio of population-scaled recombination rates on the Z and the autosomes in the 330 ZZ males from the decay of σ_D^2 with distance (86 autosomal and nine Z windows of 2 Mb, 95% CI from 200 bootstrap resamples of windows). Bottom, the ratio R of Z to autosomal synonymous divergence between *Illex illecebrosus* and *S. oualaniensis* and the male-to-female mutation ratio $\alpha = (3R - 2)/(4 - 3R)$ (Miyata et al., 1987), for all genes and with autosomal genes matched to the Z genes by d_N/d_S, by third-position GC content (GC3) or by both, or restricted to genes above the autosomal median d_N/d_S (95% CI from 2,000 bootstrap resamples of genes within strata).

Linkage disequilibrium		ρ_Z/ρ_A	95% CI
Ohta–Kimura decay fit		1.04	0.88–1.28
Curve rescaling		1.08	0.78–1.67
Synonymous divergence	**Z / autosomal genes**	**R (95% CI)**	**α (95% CI)**
All genes	109 / 8,649	0.83 (0.72–0.94)	0.33 (0.10–0.70)
Matched on d_N/d_S	109 / 8,649	0.84 (0.73–0.96)	0.36 (0.11–0.79)
Above median d_N/d_S	44 / 4,324	0.79 (0.66–0.93)	0.24 (−0.01–0.65)
Matched on GC3	109 / 8,649	0.87 (0.76–0.98)	0.43 (0.17–0.89)
Matched on d_N/d_S and GC3	109 / 6,042	0.89 (0.78–1.01)	0.49 (0.20–1.06)